

Viseme ID	IPA	Mouth position
8	o	
9	aʊ	
10	ɔɪ	
11	aɪ	

Viseme ID	IPA	Mouth position
12	h	
13	ɹ	
14	l	
15	s, z	

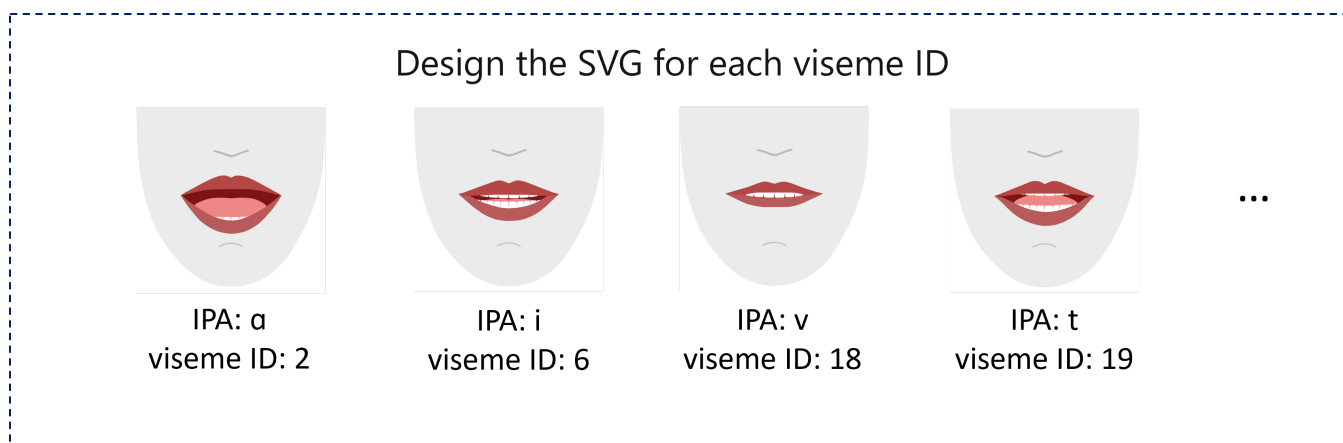
Viseme ID	IPA	Mouth position
16	ʃ, tʃ, dʒ, ʒ	
17	ð	
18	f, v	
19	d, t, n, θ	

Viseme ID	IPA	Mouth position
20	k, g, ŋ	
21	p, b, m	

2D SVG animation

For 2D characters, you can design a character that suits your scenario and use Scalable Vector Graphics (SVG) for each viseme ID to get a time-based face position.

With temporal tags that are provided in a viseme event, these well-designed SVGs is processed with smoothing modifications, and provide robust animation to the users. For example, the following illustration shows a red-lipped character designed for language learning.



3D blend shapes animation

You can use blend shapes to drive the facial movements of a 3D character that you designed.

The blend shapes JSON string is represented as a 2-dimensional matrix. Each row represents a frame. Each frame (in 60 FPS) contains an array of 55 facial positions.

Get viseme events with the Speech SDK

To get viseme with your synthesized speech, subscribe to the `VisemeReceived` event in the Speech SDK.

ⓘ Note

To request SVG or blend shapes output, you should use the `mstts:viseme` element in SSML. For details, see [how to use viseme element in SSML](#).

The following snippet shows how to subscribe to the viseme event:

C#

```
using (var synthesizer = new SpeechSynthesizer(speechConfig, audioConfig))
{
    // Subscribes to viseme received event
    synthesizer.VisemeReceived += (s, e) =>
    {
        Console.WriteLine($"Viseme event received. Audio offset: " +
            $"{e.AudioOffset / 10000}ms, viseme id: {e.VisemeId}.");

        // `Animation` is an xml string for SVG or a json string for blend shapes
        var animation = e.Animation;
    };

    // If VisemeID is the only thing you want, you can also use `SpeakTextAsync()`
    var result = await synthesizer.SpeakSsmlAsync(ssml);
}
```

Here's an example of the viseme output.

Viseme ID

text

(Viseme), Viseme ID: 1, Audio offset: 200ms.

(Viseme), Viseme ID: 5, Audio offset: 850ms.

.....

(Viseme), Viseme ID: 13, Audio offset: 2350ms.

After you obtain the viseme output, you can use these events to drive character animation. You can build your own characters and automatically animate them.

Next steps

- [SSML phonetic alphabets](#)
- [How to improve synthesis with SSML](#)

What are high definition voices?

Azure AI Speech continues to advance in the field of text to speech technology with the introduction of neural text to speech high definition (HD) voices. The HD voices can understand the content, automatically detect emotions in the input text, and adjust the speaking tone in real-time to match the sentiment. HD voices maintain a consistent voice persona from their neural (and non HD) counterparts, and deliver even more value through enhanced features.

Key features of neural text to speech HD voices

The following are the key features of Azure AI Speech HD voices:

 Expand table

Key features	Description
Human-like speech generation	Neural text to speech HD voices can generate highly natural and human-like speech. The model is trained on millions of hours of multilingual data, enabling it to accurately interpret input text and generate speech with the appropriate emotion, pace, and rhythm without manual adjustments.
Conversational	Neural text to speech HD voices can replicate natural speech patterns, including spontaneous pauses and emphasis. When given conversational text, the model can reproduce common phonemes like pauses and filler words. The generated voice sounds as if someone is conversing directly with you.
Prosody variations	Neural text to speech HD voices introduce slight variations in each output to enhance realism. These variations make the speech sound more natural, as human voices naturally exhibit variation.
High fidelity	The primary objective of neural text to speech HD voices is to generate high-fidelity audio. The synthetic speech produced by our system can closely mimic human speech in both quality and naturalness.

Comparison of Azure AI Speech HD voices to other Azure text to speech voices

How do Azure AI Speech HD voices compare to other Azure text to speech voices? How do they differ in terms of features and capabilities?

Here's a comparison of features between Azure AI Speech HD voices, Azure OpenAI HD voices, and Azure AI Speech voices:

Feature	Azure AI Speech HD voices	Azure OpenAI HD voices	Azure AI Speech voices (not HD)
Region	See Speech service regions	See Speech service regions	Available in dozens of regions. See the Speech service regions .
Number of voices	30	6	More than 500
Multilingual	Yes	Yes	Yes (applicable only to multilingual voices)
SSML support	Support for a subset of SSML elements .	Support for a subset of SSML elements .	Support for the full set of SSML in Azure AI Speech.
Development options	Speech SDK, Speech CLI, REST API	Speech SDK, Speech CLI, REST API	Speech SDK, Speech CLI, REST API
Deployment options	Cloud only	Cloud only	Cloud, embedded, hybrid, and containers.
Real-time or batch synthesis	Real-time only	Real-time and batch synthesis	Real-time and batch synthesis
Latency	Less than 300 ms	Greater than 500 ms	Less than 300 ms
Sample rate of synthesized audio	8, 16, 24, and 48 kHz	8, 16, 24, and 48 kHz	8, 16, 24, and 48 kHz
Speech output audio format	opus, mp3, pcm, truesilk	opus, mp3, pcm, truesilk	opus, mp3, pcm, truesilk

Supported Azure AI Speech HD voices

The Azure AI Speech HD voice values are in the format `voicename:basemodel:version`. The name before the colon, such as `en-US-Ava`, is the voice persona name and its original locale. The base model is tracked by versions in subsequent updates.

Currently, `DragonHD` is the only base model available for Azure AI Speech HD voices. To ensure that you're using the latest version of the base model that we provide without having to make a code change, use the `LatestNeural` version.

For example, for the persona `en-US-Ava` you can specify the following HD voice values:

- `en-US-Ava:DragonHDLatestNeural`: Always uses the latest version of the base model that we provide later.

The following table lists the Azure AI Speech HD voices that are currently available.

 Expand table

Voice Name	Gender	Status	Note
de-DE-Florian:DragonHDLatestNeural	Male	GA	
de-DE-Seraphina:DragonHDLatestNeural	Female	GA	
en-US-Adam:DragonHDLatestNeural	Male	GA	
en-US-Alloy:DragonHDLatestNeural	Male	Preview	
en-US-Andrew:DragonHDLatestNeural	Male	GA	
en-US-Andrew2:DragonHDLatestNeural	Male	GA	Optimized for conversational content
en-US-Andrew3:DragonHDLatestNeural	Male	Preview	Optimized for podcast content
en-US-Aria:DragonHDLatestNeural	Female	Preview	
en-US-Ava:DragonHDLatestNeural	Female	GA	
en-US-Ava3:DragonHDLatestNeural	Female	Preview	Optimized for podcast content
en-US-Brian:DragonHDLatestNeural	Male	GA	
en-US-Davis:DragonHDLatestNeural	Male	GA	
en-US-Emma:DragonHDLatestNeural	Female	GA	
en-US-Emma2:DragonHDLatestNeural	Female	GA	Optimized for conversational content
en-US-Jenny:DragonHDLatestNeural	Female	Preview	
en-US-MultiTalker-Ava-Andrew:DragonHDLatestNeural	Male	Preview	
en-US-Nova:DragonHDLatestNeural	Female	Preview	
en-US-Phoebe:DragonHDLatestNeural	Female	Preview	
en-US-Serena:DragonHDLatestNeural	Female	Preview	
en-US-Steffan:DragonHDLatestNeural	Male	GA	
es-ES-Tristan:DragonHDLatestNeural	Male	GA	
es-ES-Ximena:DragonHDLatestNeural	Female	GA	

Voice Name	Gender	Status	Note
fr-FR-Remy:DragonHDLatestNeural	Male	GA	
fr-FR-Vivienne:DragonHDLatestNeural	Female	GA	
ja-JP-Masaru:DragonHDLatestNeural	Male	GA	
ja-JP-Nanami:DragonHDLatestNeural	Female	GA	
zh-CN-Xiaochen:DragonHDLatestNeural	Female	GA	
zh-CN-Yunfan:DragonHDLatestNeural	Male	GA	

How to use Azure AI Speech HD voices

You can use HD voices with the same Speech SDK and REST APIs as the non HD voices.

Here are some key points to consider when using Azure AI Speech HD voices:

- **Voice locale:** The locale in the voice name indicates its original language and region.
- **Base models:**
 - HD voices come with a base model that understands the input text and predicts the speaking pattern accordingly. You can specify the desired model (such as DragonHDLatestNeural) according to the availability of each voice.
- **SSML usage:** To reference a voice in SSML, use the format `voicename:basemodel:version`. The name before the colon, such as `de-DE-Seraphina`, is the voice persona name and its original locale. The base model is tracked by versions in subsequent updates.
- **Temperature parameter:**
 - The temperature value is a float ranging from 0 to 1, influencing the randomness of the output. You can also adjust the temperature parameter to control the variation of outputs. Less randomness yields more stable results, while more randomness offers variety but less consistency.
 - Lower temperature results in less randomness, leading to more predictable outputs. Higher temperature increases randomness, allowing for more diverse outputs. The default temperature is set at 1.0.

Here's an example of how to use Azure AI Speech HD voices in SSML:

ssml

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis'
xmlns:mstts='https://www.w3.org/2001/mstts' xml:lang='en-US'>
<voice name='en-US-Ava:DragonHDLatestNeural' parameters='temperature=0.8'>Here is a
```

```
test</voice>
</speak>
```

Supported and unsupported SSML elements for Azure AI Speech HD voices

The Speech Synthesis Markup Language (SSML) with input text determines the structure, content, and other characteristics of the text to speech output. For example, you can use SSML to define a paragraph, a sentence, a break or a pause, or silence. You can wrap text with event tags such as bookmark or viseme that your application processes later.

The Azure AI Speech HD voices don't support all SSML elements or events that other Azure AI Speech voices support. Of particular note, Azure AI Speech HD voices don't support [word boundary events](#).

For detailed information on the supported and unsupported SSML elements for Azure AI Speech HD voices, refer to the following table. For instructions on how to use SSML elements, refer to the [Speech Synthesis Markup Language \(SSML\) documentation](#).

 Expand table

SSML element	Description	Supported in Azure AI Speech HD voices
<voice>	Specifies the voice and optional effects (<code>eq_car</code> and <code>eq_telecomhp8k</code>).	Yes
<mstts:express-as>	Specifies speaking styles and roles.	No
<mstts:ttseembedding>	Specifies the <code>speakerProfileId</code> property for a personal voice.	No
<lang xml:lang>	Specifies the speaking language.	Yes
<prosody>	Adjusts pitch, contour, range, rate, and volume.	No
<emphasis>	Adds or removes word-level stress for the text.	No
<audio>	Embeds prerecorded audio into an SSML document.	No
<mstts:audioduration>	Specifies the duration of the output audio.	No
<mstts:backgroundaudio>	Adds background audio to your SSML documents or mixes an audio file with text to speech.	No

SSML element	Description	Supported in Azure AI Speech HD voices
<code><phoneme></code>	Specifies phonetic pronunciation in SSML documents.	No
<code><lexicon></code>	Defines how multiple entities are read in SSML.	Yes (only supports alias)
<code><say-as></code>	Indicates the content type, such as number or date, of the element's text.	Yes
<code><sub></code>	Indicates that the alias attribute's text value should be pronounced instead of the element's enclosed text.	Yes
<code><math></code>	Uses the MathML as input text to properly pronounce mathematical notations in the output audio.	No
<code><bookmark></code>	Gets the offset of each marker in the audio stream.	No
<code><break></code>	Overrides the default behavior of breaks or pauses between words.	No
<code><mstts:silence></code>	Inserts pause before or after text, or between two adjacent sentences.	No
<code><mstts:viseme></code>	Defines the position of the face and mouth while a person is speaking.	No
<code><p></code>	Denotes paragraphs in SSML documents.	Yes
<code><s></code>	Denotes sentences in SSML documents.	Yes

ⓘ Note

Although a [previous section in this guide](#) also compared Azure AI Speech HD voices to Azure OpenAI HD voices, the SSML elements supported by Azure AI Speech aren't applicable to Azure OpenAI voices.

HD Flash voices

HD voices are currently supported in `eastus`, `westeurope`, and `southeastasia` regions, to provide similar capabilities for customers in China regions (`chinaeast2`, `chinanorth2`, `chinanorth3`), we offer HDFlash versions of selected HD voices. These HDFlash voices deliver

enhanced naturalness compared to standard voices. You also can find those HDFlash voices in `eastus`, `westeurope`, and `southeastasia`.

Below is the complete list of available HD Flash voices:

 Expand table

Voice Name	Gender
zh-CN-Xiaochen:DragonHDFlashLatestNeural	Female
zh-CN-Xiaoxiao:DragonHDFlashLatestNeural	Female
zh-CN-Xiaoxiao2:DragonHDFlashLatestNeural	Female
zh-CN-Yunxia:DragonHDFlashLatestNeural	Male
zh-CN-Yunxiao:DragonHDFlashLatestNeural	Male
zh-CN-Yunye:DragonHDFlashLatestNeural	Male
zh-CN-Yunyi:DragonHDFlashLatestNeural	Male

 Note

HD Flash only support text in `zh-CN` and `en-US`.

Related content

- [Try the text to speech quickstart in Azure AI Speech](#)
- [Learn more about how to use SSML and events](#)

Last updated on 11/06/2025

What is voice conversion? (Preview)

07/10/2025

 Note

This feature is currently in public preview. This preview is provided without a service-level agreement, and is not recommended for production workloads. Certain features might not be supported or might have constrained capabilities. For more information, see [Supplemental Terms of Use for Microsoft Azure Previews](#).

Voice conversion is the process of transforming the voice characteristics of a given audio to a target voice speaker. After voice conversion, the resulting audio reserves source audio's linguistic content and prosody while the voice timbre sounds like the target speaker.

There are 3 reasons users need voice conversion functionality:

- Voice conversion can replicate your content using a different voice identity while maintaining the original prosody and emotion. For instance, in education, teachers can record themselves reading stories, and voice conversion can deliver these stories using a pre-designed cartoon character's voice. This method preserves the expressiveness of the teacher's reading while incorporating the unique timbre of the cartoon character's voice.
- Another application is multilingual dubbing. When localized content is read by different voices, voice conversion can transform them into a uniform voice, ensuring a consistent experience across all languages while keeping the most localized voice characters.
- Voice conversion enhances the control over the expressiveness of a voice. By transforming various speaking styles, such as adopting a unique tone or conveying exaggerated emotions, a voice gains greater versatility in expression and can be more dynamic in different scenarios.

Key capabilities

Voice conversion (or voice changer or speech to speech conversion) is built on state-of-the-art generative models and offers high-quality voice conversion. It delivers the following core capabilities:

 Expand table

Capability	Description
High speaker similarity	Captures the timbre and vocal identity of the target speaker.

Capability	Description
	Generates audio that accurately matches the target voice.
Prosody preservation	Maintains rhythm, stress, and intonation of source audio. Preserves expressive and emotional qualities.
High audio fidelity	Generates realistic, natural-sounding audio. Minimizes artifacts.
Multilingual support	Enables multilingual voice conversion. Supports 91 locales (same as standard text to speech locale support). See supported voices for voice conversion for the complete list.

Use voice conversion

You can use Azure AI Speech voice conversion with either the Speech SDK or text to speech REST APIs.

Use the `<mstts:voiceconversion>` tag via Speech Synthesis Markup Language (SSML) to specify the source audio URL and the target voice for the conversion. For a complete list of supported target voices, see [supported voices for voice conversion](#).

Example SSML

XML

```
< speak version="1.0" xmlns="http://www.w3.org/2001/10/synthesis"
xmlns:mstts="https://www.w3.org/2001/mstts" xml:lang="en-US">
  < voice xml:lang="en-US" xml:gender="Female" name="en-US-AvaMultilingualNeural">
    <mstts:voiceconversion
url="https://your.blob.core.windows.net/sourceaudio.wav"/>
  </voice>
</speak>
```

For details about the SSML structure and usage, see the [Speech Synthesis Markup Language \(SSML\) reference](#) documentation.

Related content

- [Text to speech overview](#)
- [Speech synthesis markup voice](#)

What is custom voice?

08/07/2025

Custom voice is a text to speech feature that lets you create a one-of-a-kind, customized, synthetic voice for your applications. With custom voice, you can build a highly natural-sounding voice for your brand or characters by providing human speech samples as fine-tuning data.

Important

Custom voice access is [limited](#) based on eligibility and usage criteria. Request access on the [intake form](#) .

Out of the box, [text to speech](#) can be used with standard voices for each [supported language](#). The standard voices work well in most text to speech scenarios if a unique voice isn't required.

Custom voice is based on the neural text to speech technology and the multilingual, multi-speaker, universal model. You can create synthetic voices that are rich in speaking styles, or adaptable cross languages. The realistic and natural sounding voice of custom voice can represent brands, personify machines, and allow users to interact with applications conversationally. See the [supported languages](#) for custom voice.

How does it work?

To create a custom voice, use [Speech Studio](#)  to upload the recorded audio and corresponding scripts, train the model, and deploy the voice to a custom endpoint.

Creating a great custom voice requires careful quality control in each step, from voice design and data preparation, to the deployment of the voice model to your system.

Before you get started in Speech Studio, here are some considerations:

- [Design a persona](#) of the voice that represents your brand by using a persona brief document. This document defines elements such as the features of the voice, and the character behind the voice. This helps to guide the process of creating a custom voice model, including defining the scripts, selecting your voice talent, training, and voice tuning.
- [Select the recording script](#) to represent the user scenarios for your voice. For example, you can use the phrases from bot conversations as your recording script if you're creating a customer service bot. Include different sentence types in your scripts, including statements, questions, and exclamations.

Here's an overview of the steps to create a custom voice in Speech Studio:

1. [Create a project](#) to contain your data, voice models, tests, and endpoints. Each project is specific to a country/region and language. If you're going to create multiple voices, it's recommended that you create a project for each voice.
2. [Set up voice talent](#). Before you can fine-tune a professional voice, you must submit a recording of the voice talent's consent statement. The voice talent statement is a recording of the voice talent reading a statement that they consent to the usage of their speech data for professional voice fine-tuning.
3. [Prepare fine-tuning data](#) in the right [format](#). It's a good idea to capture the audio recordings in a professional quality recording studio to achieve a high signal-to-noise ratio. The quality of the voice model depends heavily on your fine-tuning data. Consistent volume, speaking rate, pitch, and consistency in expressive mannerisms of speech are required.
4. [Train your voice model](#). Select at least 300 utterances to create a custom voice. A series of data quality checks are automatically performed when you upload them. To build high-quality voice models, you should fix any errors and submit again.
5. [Test your voice](#). Prepare test scripts for your voice model that cover the different use cases for your apps. It's a good idea to use scripts within and outside the training dataset, so you can test the quality more broadly for different content.
6. [Deploy and use your voice model](#) in your apps.

You can tune, adjust, and use your custom voice, similarly as you would use a standard voice. Convert text into speech in real-time, or generate audio content offline with text input. You use the [REST API](#), the [Speech SDK](#), or the [Speech Studio](#) [↗](#).

Tip

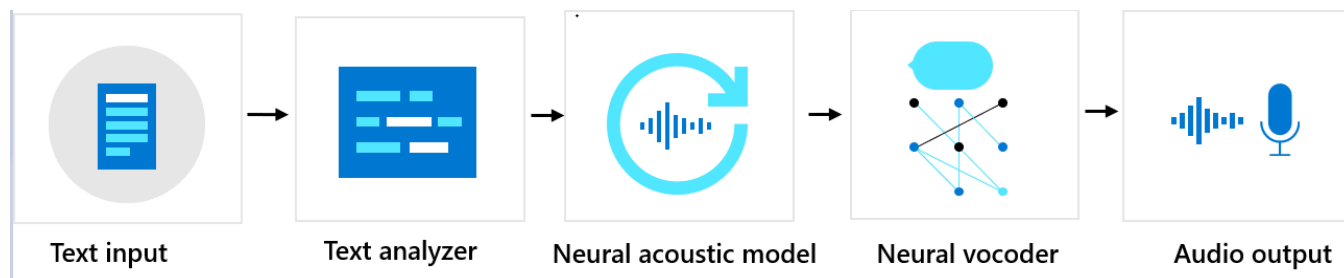
Check out the code samples in the [Speech SDK repository on GitHub](#) [↗](#) to see how to use custom voice in your application.

The style and the characteristics of the trained voice model depend on the style and the quality of the recordings from the voice talent used for training. However, you can make several adjustments by using [SSML \(Speech Synthesis Markup Language\)](#) when you make the API calls to your voice model to generate synthetic speech. SSML is the markup language used to communicate with the text to speech service to convert text into audio. The adjustments you can make include change of pitch, rate, intonation, and pronunciation correction. If the voice model is built with multiple styles, you can also use SSML to switch the styles.

Components sequence

Custom voice consists of three major components: the text analyzer, the neural acoustic model, and the neural vocoder. To generate natural synthetic speech from text, text is first input into the text analyzer, which provides output in the form of phoneme sequence. A *phoneme* is a basic unit of sound that distinguishes one word from another in a particular language. A sequence of phonemes defines the pronunciations of the words provided in the text.

Next, the phoneme sequence goes into the neural acoustic model to predict acoustic features that define speech signals. Acoustic features include the timbre, the speaking style, speed, intonations, and stress patterns. Finally, the neural vocoder converts the acoustic features into audible waves, so that synthetic speech is generated.



Neural text to speech voice models are trained by using deep neural networks based on the recording samples of human voices. For more information, see [this Microsoft blog post](#). To learn more about how a neural vocoder is trained, see [this Microsoft blog post](#).

Responsible AI

An AI system includes not only the technology, but also the people who use it, the people who are affected by it, and the environment in which it's deployed. Read the transparency notes to learn about responsible AI use and deployment in your systems.

- [Transparency note and use cases for custom voice](#)
- [Characteristics and limitations for using custom voice](#)
- [Limited access to custom voice](#)
- [Guidelines for responsible deployment of synthetic voice technology](#)
- [Disclosure for voice talent](#)
- [Disclosure design guidelines](#)
- [Disclosure design patterns](#)
- [Code of Conduct for Text to speech integrations](#)
- [Data, privacy, and security for custom voice](#)

Next steps

- [Create a project](#)
- [Prepare fine-tuning data](#)

- Train model

Professional voice fine-tuning data

08/07/2025

When you're ready to create a custom voice for your application, the first step is to gather audio recordings and associated scripts to start professional voice fine-tuning. "Custom voice" is an umbrella term that includes both professional voice fine-tuning and personal voice. The Speech service uses this data for professional voice fine-tuning, creating a unique voice tuned to match the voice in the recordings. After you fine-tune a professional voice, you can start synthesizing speech in your applications.



Tip

To create a voice for production use, we recommend you use a professional recording studio and voice talent. For more information, see [record voice samples for professional voice fine-tuning](#).

Types of data for professional voice fine-tuning

A dataset for professional voice fine-tuning includes audio recordings and a text file with the associated transcriptions. Each audio file should contain a single utterance (a single sentence or a single turn for a dialog system), and be less than 15 seconds long.

In some cases, you might not have the right dataset ready. You can test professional voice fine-tuning with available audio files, short or long, with or without transcripts.

This table lists data types and how each is used for professional voice fine-tuning.

Expand table

Data type	Description	When to use	Extra processing required	Processed as
Individual utterances + matching transcript	A collection (.zip) of audio files (.wav) as individual utterances. Each audio file should be 15 seconds or less in length, paired with a formatted transcript (.txt).	Professional recordings with matching transcripts	Ready for fine-tuning.	Segmented
Long audio + transcript	A collection (.zip) of long, unsegmented audio files (.wav or .mp3, longer than 20 seconds, at most 1,000 audio	You have audio files and matching transcripts, but they aren't	Segmentation (using batch transcription). Audio format	Segmented, Contextual

Data type	Description	When to use	Extra processing required	Processed as
	files), paired with a collection (.zip) of transcripts that contains all spoken words.	segmented into utterances.	transformation wherever required.	
Audio only (Preview)	A collection (.zip) of audio files (.wav or .mp3, at most 1,000 audio files) without a transcript.	You only have audio files available, without transcripts.	Segmentation + transcript generation (using batch transcription). Audio format transformation wherever required.	Segmented, Contextual

Files should be grouped by type into a dataset and uploaded as a zip file. Each dataset can only contain a single data type.

ⓘ Note

The maximum number of datasets allowed to be imported per subscription is 500 zip files for standard subscription (S0) users.

Processed as Contextual would retain the audio as a whole to keep the contextual information for more natural intonations.

Individual utterances + matching transcript

You can prepare recordings of individual utterances and the matching transcript in two ways. Either [write a script and have it read by a voice talent](#) or use publicly available audio and transcribe it to text. If you do the latter, edit disfluencies from the audio files, such as "um" and other filler sounds, stutters, mumbled words, or mispronunciations.

To produce a good voice model, create the recordings in a quiet room with a high-quality microphone. Consistent volume, speaking rate, speaking pitch, and expressive mannerisms of speech are essential.

For data format examples, refer to the sample dataset on [GitHub](#). The sample dataset includes the sample script and the associated audio.

Audio data for Individual utterances + matching transcript

Each audio file should contain a single utterance (a single sentence or a single turn of a dialog system), less than 15 seconds long. All files must be in the same spoken language. Multi-language custom Text to speech voices aren't supported, except for the Chinese-English bilingual. Each audio file must have a unique filename with the filename extension .wav.

Follow these guidelines when preparing audio.

 Expand table

Property	Value
File format	RIFF (.wav), grouped into a .zip file
File name	File name characters supported by Windows OS, with .wav extension. The characters <code>\ / : * ? " < > \ </code> aren't allowed. It can't start or end with a space, and can't start with a dot. No duplicate file names allowed.
Sampling rate	24 KHz and higher required when fine-tuning a professional voice.
Sample format	PCM, at least 16-bit
Audio length	Shorter than 15 seconds
Archive format	.zip
Maximum archive size	2048 MB

 Note

The default sampling rate for professional voice fine-tuning is 24 KHz. Audio files with a sampling rate lower than 16,000 Hz will be rejected. If a .zip file contains .wav files with different sample rates, only those equal to or higher than 16,000 Hz will be imported. Your audio files with a sampling rate higher than 16,000 Hz and lower than 24 KHz will be up-sampled to 24 KHz for fine-tuning. It's recommended that you use a sample rate of 24 KHz and higher for your fine-tuning data.

Transcription data for Individual utterances + matching transcript

The transcription file is a plain text file. Use these guidelines to prepare your transcriptions.

 Expand table

Property	Value
File format	Plain text (.txt)
Encoding format	ANSI, ASCII, UTF-8, UTF-8-BOM, UTF-16-LE, or UTF-16-BE. For zh-CN, ANSI and ASCII encoding aren't supported.
# of utterances per line	One - Each line of the transcription file should contain the name of one of the audio files, followed by the corresponding transcription. You must use a tab (\t) to separate the file name and transcription.
Maximum file size	2048 MB

Here's an example of how the transcripts are organized utterance by utterance in one .txt file:

```
0000000001[tab] This is the waistline, and it's falling.
0000000002[tab] We have trouble scoring.
0000000003[tab] It was Janet Maslin.
```

It's important that the transcripts are 100% accurate transcriptions of the corresponding audio. Errors in the transcripts introduce quality loss during the fine-tuning process.

Long audio + transcript (Preview)

ⓘ Note

For **Long audio + transcript (Preview)**, only these languages are supported: Chinese (Mandarin, Simplified), Chinese (Cantonese, Traditional), Chinese (Taiwanese Mandarin), English (India), English (United Kingdom), English (United States), French (France), German (Germany), Hindi (India), Italian (Italy), Japanese (Japan), Portuguese (Brazil), Spanish (Spain) and Spanish (Mexico).

Processed as Contextual is currently only available for Chinese (Mandarin, Simplified) and English (United States).

In some cases, you might not have segmented audio available. The Speech Studio can help you segment long audio files and create transcriptions. The long-audio segmentation service uses the [Batch Transcription API](#) feature of speech to text.

The service offers two processing modes:

- **Segmented:** The default processing mode that works with all supported languages
- **Contextual:** An enhanced mode that retains the audio as a whole to keep the contextual information for more natural intonations.

During the processing of the segmentation, your audio files and the transcripts are also sent to the custom speech service to refine the recognition model so the accuracy can be improved for your data. No data is retained during this process. After the segmentation is done, only the utterances segmented and their mapping transcripts will be stored for your downloading and fine-tuning.

Audio data for Long audio + transcript

Follow these guidelines when preparing audio for segmentation.

 Expand table

Property	Value
File format	RIFF (.wav) or .mp3, grouped into a .zip file
File name	File name characters supported by Windows OS, with .wav extension. The characters <code>\ / : * ? " < > \ </code> aren't allowed. It can't start or end with a space, and can't start with a dot. No duplicate file names allowed.
Sampling rate	24 KHz and higher required when fine-tuning a professional voice.
Sample format	RIFF(.wav): PCM, at least 16-bit. mp3: At least 256 Kbps bit rate.
Audio length	Longer than 30 seconds
Archive format	.zip
Maximum archive size	2048 MB, at most 1,000 audio files included

Note

The default sampling rate for professional voice fine-tuning is 24 KHz. Audio files with a sampling rate lower than 16,000 Hz will be rejected. Your audio files with a sampling rate higher than 16,000 Hz and lower than 24 KHz will be up-sampled to 24 KHz for fine-tuning. It's recommended that you should use a sample rate of 24 KHz and higher for your fine-tuning data.

Segmented utterances should ideally be between 5 and 15 seconds long. For optimal segmentation results, it is recommended to include natural pauses of 0.5 to 1 second every 5 to 15 seconds of speech, preferably at the end of phrases or sentences.

All audio files should be grouped into a zip file. It's OK to put .wav files and .mp3 files into the same zip file. For example, you can upload a 45-second audio file named 'kingstory.wav' and a 200-second long audio file named 'queenstory.mp3' in the same zip file. All .mp3 files will be transformed into the .wav format after processing.

Transcription data for Long audio + transcript

Transcripts must be prepared to the specifications listed in this table. Each audio file must be matched with a transcript.

 Expand table

Property	Value
File format	Plain text (.txt), grouped into a .zip
File name	Use the same name as the matching audio file
Encoding format	ANSI, ASCII, UTF-8, UTF-8-BOM, UTF-16-LE, or UTF-16-BE. For zh-CN, ANSI and ASCII encoding aren't supported.
# of utterances per line	No limit
Maximum file size	2048 MB

All transcripts files in this data type should be grouped into a zip file. For example, you might upload a 45-second audio file named 'kingstory.wav' and a 200-second long audio file named 'queenstory.mp3' in the same zip file. You need to upload another zip file containing the corresponding two transcripts--one named 'kingstory.txt' and the other one named 'queenstory.txt'. Within each plain text file, you provide the full correct transcription for the matching audio.

After your dataset is successfully uploaded, we'll help you segment the audio file into utterances based on the transcript provided. You can check the segmented utterances and the matching transcripts by downloading the dataset. Unique IDs are assigned to the segmented utterances automatically. It's important that you make sure the transcripts you provide are 100% accurate. Errors in the transcripts can reduce the accuracy during the audio segmentation and further introduce quality loss in the fine-tuning phase that comes later.

Audio only (Preview)

ⓘ Note

For **Audio only (Preview)**, only these languages are supported: Chinese (Mandarin, Simplified), Chinese (Cantonese, Traditional), Chinese (Taiwanese Mandarin), English (India), English (United Kingdom), English (United States), French (France), German (Germany), Hindi (India), Italian (Italy), Japanese (Japan), Portuguese (Brazil), Spanish (Spain) and Spanish (Mexico).

Processed as Contextual is currently only available for Chinese (Mandarin, Simplified) and English (United States).

If you don't have transcriptions for your audio recordings, use the **Audio only** option to upload your data. Our system can help you segment and transcribe your audio files.

The service offers two processing modes:

- **Segmented:** The default processing mode that works with all supported languages
- **Contextual:** An enhanced mode that retains the audio as a whole to keep the contextual information for more natural intonations.

Follow these guidelines when preparing audio.

 Expand table

Property	Value
File format	RIFF (.wav) or .mp3, grouped into a .zip file
File name	File name characters supported by Windows OS, with .wav extension. The characters <code>\ / : * ? " < > \ </code> aren't allowed. It can't start or end with a space, and can't start with a dot. No duplicate file names allowed.
Sampling rate	24 KHz and higher required when fine-tuning a professional voice.
Sample format	RIFF(.wav): PCM, at least 16-bit mp3: At least 256 Kbps bit rate.
Audio length	No limit
Archive format	.zip
Maximum archive size	2048 MB, at most 1,000 audio files included

ⓘ Note

The default sampling rate for professional voice fine-tuning is 24 KHz. Your audio files with a sampling rate higher than 16,000 Hz and lower than 24 KHz will be up-sampled to 24 KHz for fine-tuning. It's recommended that you should use a sample rate of 24 KHz and higher for your fine-tuning data.

Segmented utterances should ideally be between 5 and 15 seconds long. For optimal segmentation results, it is recommended to include natural pauses of 0.5 to 1 second every 5 to 15 seconds of speech, preferably at the end of phrases or sentences.

All audio files should be grouped into a zip file. Once your dataset is successfully uploaded, the Speech service helps you segment the audio file into utterances based on our speech batch transcription service. You can select either the Standard or Contextual processing mode, depending on your language and requirements. Unique IDs are assigned to the segmented utterances automatically. Matching transcripts are generated through speech recognition. All .mp3 files will be transformed into the .wav format after processing. You can check the segmented utterances and the matching transcripts by downloading the dataset.

Next steps

- [Train your voice model](#)
- [Deploy and use your voice model](#)
- [How to record voice samples](#)

Recording voice samples for custom voice

08/07/2025

This article provides you with best practices on preparing high-quality voice samples for professional voice fine-tuning. To understand how the data is processed and the minimum requirements for data acceptance, please refer to [upload your data](#).

Creating a high-quality professional voice from scratch isn't a casual undertaking. The central component of a custom voice is a large collection of audio samples of human speech. It's vital that these audio recordings be of high quality. Choose a voice talent who has experience making these kinds of recordings, and have them recorded by a recording engineer using professional equipment.

Before you can make these recordings, though, you need a script: the words are spoken by your voice talent to create the audio samples.

Many small but important details go into creating a professional voice recording. This guide is a roadmap for a process that will help you get good, consistent results.

Tips for preparing data for a high-quality voice

A highly natural custom voice depends on several factors, like the quality and size of your training data.

The quality of your training data is a primary factor. For example, in the same training set, consistent volume, speaking rate, speaking pitch, and speaking style are essential to create a high-quality custom voice. You should also avoid background noise in the recording and make sure the script and recording match. To ensure the quality of your data, you need to follow [script selection criteria](#) and [recording requirements](#).

Regarding the size of the training data, in most cases you can build a reasonable custom voice with 300 utterances. According to our tests, adding more training data in most languages doesn't necessarily improve naturalness of the voice itself (tested using the MOS score), however, with more training data that covers more word instances, you have higher possibility to reduce the ratio of dissatisfactory parts of speech for the voice, such as the glitches. To hear what dissatisfactory parts of speech sound like, refer to [the GitHub examples](#).

In some cases, you might want a voice persona with unique characteristics. For example, a cartoon persona needs a voice with a special speaking style, or a voice that is dynamic in intonation. For such cases, we recommend that you prepare at least 1000 (preferably 2000) utterances, and record them at a professional recording studio. To learn more about how to

improve the quality of your voice model, see [characteristics and limitations for using custom voice](#).

Voice recording roles

There are four basic roles in a custom voice recording project:

 [Expand table](#)

Role	Purpose
Voice talent	This person's voice forms the basis of the custom voice.
Recording engineer	Oversees the technical aspects of the recording and operates the recording equipment.
Director	Prepares the script and coaches the voice talent's performance.
Editor	Finalizes the audio files and prepares them for upload to the Speech service.

An individual can fill more than one role. This guide assumes that you are filling the director role and hiring both a voice talent and a recording engineer. If you want to make the recordings yourself, this article includes some information about the recording engineer role. The editor role isn't needed until after the recording session. In the meantime, the director or the recording engineer can fill this role.

Choose your voice talent

Actors with experience in voiceover, voice character work, announcing or news reading make good voice talent. Choose voice talent whose natural voice you like. It's possible to create unique "character" voices, but it's harder for most talent to perform them consistently, and the effort can cause voice strain. The single most important factor for choosing voice talent is consistency. Your recordings for the same voice style should all sound like they were made on the same day in the same room. You can approach this ideal through good recording practices and engineering.

Your voice talent must be able to speak with consistent rate, volume level, pitch, and tone with clear dictation. They also need to be able to control their pitch variation, emotional effect, and speech mannerisms. Recording voice samples can be more fatiguing than other kinds of voice work, so most voice talents can only record for two or three hours a day. Limit sessions to three or four days a week, with a day off in-between if possible.

Work with your voice talent to develop a persona that defines the overall sound and emotional tone of the custom voice. Define the speaking styles for your persona and ask your voice talent to read the script in a way that aligns with your desired styles. Ensure that the speaking style remains consistent throughout the recordings for a set of training data.

For example, a persona with a naturally upbeat personality would carry a note of optimism in their voice. However, this personality should be expressed consistently across all recordings for a set of training data. Listen to existing voices to get a sense of what you're aiming for.

Tip

Usually, you'll want to own the voice recordings you make. Your voice talent should be amenable to a work-for-hire contract for the project.

Create a script

The starting point of any custom voice recording session is the script, which contains the utterances to be spoken by your voice talent. The term "utterances" encompasses both full sentences and shorter phrases. Building a custom voice requires at least 300 recorded utterances as training data.

The utterances in your script can come from anywhere: fiction, non-fiction, transcripts of speeches, news reports, and anything else available in printed form. For a brief discussion of potential legal issues, see the "[Legalities](#)" section. You can also write your own text.

Your utterances don't need to come from the same source, the same kind of source, or have anything to do with each other. However, if you use set phrases (for example, "You have successfully logged in") in your speech application, make sure to include them in your script. It gives your custom voice a better chance of pronouncing those phrases well.

We recommend the recording scripts include both general sentences and domain-specific sentences. For example, if you plan to record 2,000 sentences, 1,000 of them could be general sentences, another 1,000 of them could be sentences from your target domain or the use case of your application.

We provide [sample scripts in the 'General', 'Chat' and 'Customer Service' domains for each language](#)  to help you prepare your recording scripts. You can use these Microsoft shared scripts for your recordings directly or use them as a reference to create your own.

Script selection criteria

Below are some general guidelines that you can follow to create a good corpus (recorded audio samples) for professional voice fine-tuning.

- For most use cases, sentences are recommended to be between 2 and 15 seconds long, containing 5 to 30 words for Latin-based languages or 4 to 80 words for non-Latin languages. Aim to balance your script to include a variety of sentence types and lengths. Ensure your script does not include any duplicate sentences.

If your use case requires a high emphasis on questions, exclamations, or a mix of particularly long and short sentences, it is recommended to include a good portion of sentences as questions or exclamations, along with very short phrases and longer phrases up to 20 seconds in length.

For how to balance the different sentence types, refer to the following table:

 Expand table

Sentence types	Coverage
Statement sentences	Statement sentences should be 70-80% of the script.
Short word/phrase	Short word/phrase scripts should be about 10% of total utterances, with 5 to 7 words per case. Short words or phrases should be separated by commas to help remind voice talent to pause briefly while reading.
Question sentences (Optional)	Question sentences should be about 10%-20% of your domain script, including 5%-10% of rising and 5%-10% of falling tones. These sentences are required if you want the generated voice to accurately convey questions.
Exclamation sentences (Optional)	Exclamation sentences should be about 10%-20% of your script. These sentences are required if you want the generated voice to accurately convey exclamations.

 Note

You can estimate the number of words in a sentence by assuming a speech rate in words per second based on your language.

Best practices include:

- Balanced coverage for Parts of Speech, like verbs, nouns, adjectives, and so on.
- Balanced coverage for pronunciations. Include all letters from A to Z so the Text to speech engine learns how to pronounce each letter in your style.
- Readable, understandable, common-sense scripts for the speaker to read.

- Avoid too many similar patterns for words/phrases, like "easy" and "easier".
- Include different formats of numbers: address, unit, phone, quantity, date, and so on, in all sentence types.
- Include spelling sentences if it's something your custom voice will read. For example, "The spelling of Apple is A P P L E".

ⓘ Note

For Contextual processing mode, which provides more natural intonations and better conversational capabilities:

- Use paragraph-level text rather than sentence-level text for recordings. This approach helps capture natural speech flow between sentences and preserves contextual information.
- Each recording should ideally be longer than 30 seconds (containing more than 60 words for Latin-based languages or 160 words for non-Latin languages).
- A contextual training set with more than 30 minutes of total audio or 300 utterances can be used for training a custom voice.

- Don't put multiple sentences into one line/one utterance. Separate each line by utterance.
- Make sure the sentence is clean. Generally, don't include too many nonstandard words like numbers or abbreviations as they're hard to read. Some applications might require the reading of many numbers or acronyms. In these cases, you can include these words, but normalize them in their spoken form.

Below are some best practices for example:

- For lines with abbreviations, instead of "BTW", write "by the way".
- For lines with digits, instead of "911", write "nine one one".
- For lines with acronyms, instead of "ABC", write "A B C".

With that, make sure your voice talent pronounces these words in an expected way. Keep your script and recordings matched during the training process.

- Your script should include many different words and sentences with different kinds of sentence lengths, structures, and moods.
- Check the script carefully for errors. If possible, have someone else check it too. When you run through the script with your voice talent, you might catch more mistakes.

Difference between voice talent script and training script

The training script can differ from the voice talent script, especially for scripts that contain digits, symbols, abbreviations, date, and time. Scripts prepared for the voice talent must follow native reading conventions, such as 50% and \$45. The scripts used for training must be normalized to match the audio recording, such as *fifty percent* and *forty-five dollars*.

 Note

We provide some example scripts for the voice talent on [GitHub](#)[↗]. To use the example scripts for training, you must normalize them according to the recordings of your voice talent before uploading the file.

The following table shows the difference between scripts for voice talent and the normalized script for training.

 Expand table

Category	Voice talent script example	Training script example (normalized)
Digits	123	one hundred and twenty-three
Symbols	50%	fifty percent
Abbreviation	ASAP	as soon as possible
Date and time	March 3rd at 5:00 PM	March third at five PM

Typical defects of a script

The script's poor quality can adversely affect the training results. To achieve high-quality training results, it's crucial to avoid defects.

Script defects generally fall into the following categories:

 Expand table

Category	Example
Meaningless content.	"Colorless green ideas sleep furiously."
Incomplete sentences.	- "This was my last eve" (no subject, no specific meaning) - "They're already funny (no quote mark in the end, it's not a complete sentence)
Typo in the sentences.	- Start with a lower case - No ending punctuation if needed

Category	Example
	- Misspelling - Lack of punctuation: no period in the end (except news title) - End with symbols, except comma, question, exclamation - Wrong format, such as: - 45\$ (should be \$45) - No space or excess space between word/punctuation
Duplication in similar format, one per each pattern is enough.	- "Now is 1pm in New York" - "Now is 2pm in New York" - "Now is 3pm in New York" - "Now is 1pm in Seattle" - "Now is 1pm in Washington D.C."
Uncommon foreign words: only commonly used foreign words are acceptable in the script.	In English one might use the French word "faux" in common speech, but a French expression such as "coincer la bulle" would be uncommon.
Emoji or any other uncommon symbols	

Script format

The script is for use during recording sessions, so you can set it up any way you find easy to work with. Create the text file that's required by Speech Studio separately.

A basic script format contains three columns:

- The number of the utterance, starting at 1. Numbering makes it easy for everyone in the studio to refer to a particular utterance ("let's try number 356 again"). You can use the Microsoft Word paragraph numbering feature to number the rows of the table automatically.
- A blank column where you write the take number or time code of each utterance to help you find it in the finished recording.
- The text of the utterance itself.



Time Code	Utterance
1	Four score and seven years ago our fathers brought forth on this continent a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal.
2	Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure.
3	We are met on a great battlefield of that war.
4	We have come to dedicate a portion of that field as a final resting place for those who here gave their lives that that nation might live.

ⓘ Note

Most studios record in short segments known as "takes". Each take typically contains 10 to 24 utterances. Just noting the take number is sufficient to find an utterance later. If you're recording in a studio that prefers to make longer recordings, you'll want to note the time code instead. The studio will have a prominent time display.

Leave enough space after each row to write notes. Be sure that no utterance is split between pages. Number the pages, and print your script on one side of the paper.

Print three copies of the script: one for the voice talent, one for the recording engineer, and one for the director (you). Use a paper clip instead of staples: an experienced voice artist separates the pages to avoid making noise as the pages are turned.

Voice talent statement

To train a neural voice, you must [create a voice talent profile](#) with an audio file recorded by the voice talent consenting to the usage of their speech data to fine-tune a professional voice model. When preparing your recording script, make sure you include the statement sentence.

Legalities

Under copyright law, an actor's reading of copyrighted text might be a performance for which the author of the work should be compensated. This performance won't be recognizable in the final product, the custom voice. Even so, the legality of using a copyrighted work for this

purpose isn't well established. Microsoft can't provide legal advice on this issue; consult your own legal counsel.

Fortunately, it's possible to avoid these issues entirely. There are many sources of text you can use without permission or license.

 Expand table

Text source	Description
CMU Arctic corpus 	About 1100 sentences selected from out-of-copyright works specifically for use in speech synthesis projects. An excellent starting point.
Works no longer under copyright	Typically works published prior to 1923. For English, Project Gutenberg  offers tens of thousands of such works. You might want to focus on newer works, as the language is closer to modern English.
Government works	Works created by the United States government aren't copyrighted in the United States, though the government can claim copyright in other countries/regions.
Public domain	Works for which copyright is explicitly disclaimed or dedicated to the public domain. It might not be possible to waive copyright entirely in some jurisdictions.
Permissively licensed works	Works distributed under a license like Creative Commons or the GNU Free Documentation License (GFDL). Wikipedia uses the GFDL. Some licenses, however, may impose restrictions on performance of the licensed content that might affect the creation of a custom voice model, so read the license carefully.

Recording your script

Record your script at a professional recording studio that specializes in voice work. They have a recording booth, the right equipment, and the right people to operate it. It's recommended not to skimp on recording.

Discuss your project with the studio's recording engineer and listen to their advice. The recording should have little or no dynamic range compression (maximum of 4:1). It's critical that the audio has consistent volume and a high signal-to-noise ratio, while being free of unwanted sounds.

Recording requirements

To achieve high-quality training results, follow the following requirements during recording or data preparation:

- Clear and well pronounced

- Natural speed: not too slow or too fast between audio files.
- Appropriate volume, prosody and break: stable within the same sentence or between sentences, correct break for punctuation.
- No noise during recording
- Fit your persona design
- No wrong accent: fit to the target design
- No wrong pronunciation

You can refer to below specification to prepare for the audio samples as best practice.

 Expand table

Property	Value
File format	*.wav, Mono
Sampling rate	24 KHz
Sample format	16 bit, PCM
Peak volume levels	-3 dB to -6 dB
SNR	> 35 dB
Silence	- There should have some silence (recommend 100 ms) at the beginning and ending, but no longer than 200 ms - Silence between words or phrases < -30 dB - Silence in the wave after last word is spoken < -60 dB
Environment noise or echo	- The level of noise at start of the wave before speaking < -70 dB

Note

You can record at higher sampling rate and bit depth, for example in the format of 48 KHz 24 bit PCM. During the professional voice fine-tuning, we'll down sample it to 24 KHz 16 bit PCM automatically.

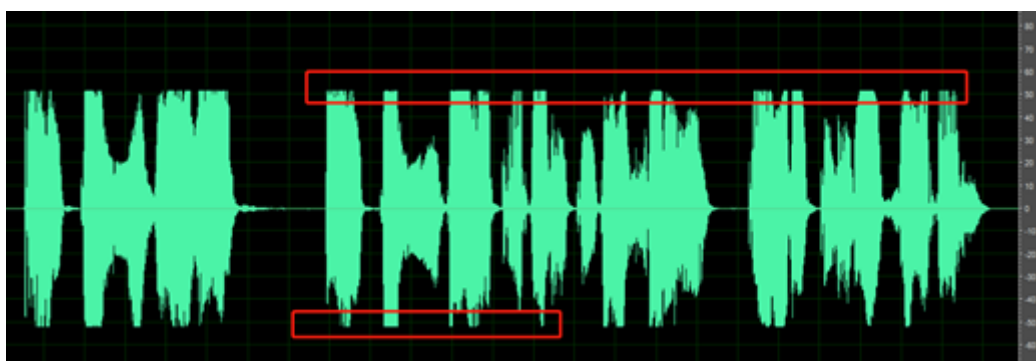
A higher signal-to-noise ratio (SNR) indicates lower noise in your audio. You can typically reach a 35+ SNR by recording at professional studios. Audio with an SNR below 20 can result in obvious noise in your generated voice.

Consider re-recording any utterances with low pronunciation scores or poor signal-to-noise ratios. If you can't re-record, consider excluding those utterances from your data.

Typical audio errors

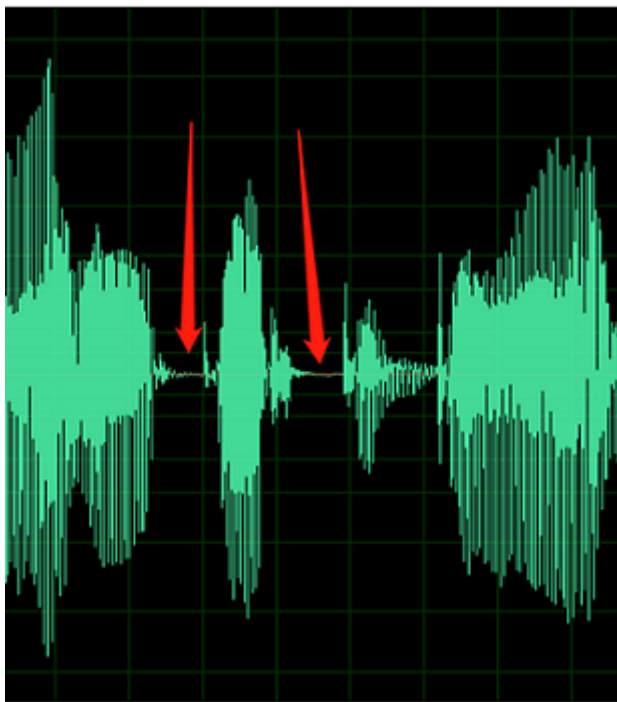
For high-quality training results, avoiding audio errors is highly recommended. Audio errors are usually within following categories:

- Audio file name doesn't match the script ID.
- WAR file has an invalid format and can't be read.
- Audio sampling rate is lower than 16 KHz. It's recommended that the .wav file sampling rate be equal or higher than 24 KHz for high-quality neural voice.
- Volume peak isn't within the range of -3 dB (70% of max volume) to -6 dB (50%).
- Waveform overflow: the waveform is cut at its peak value and is thus not complete.

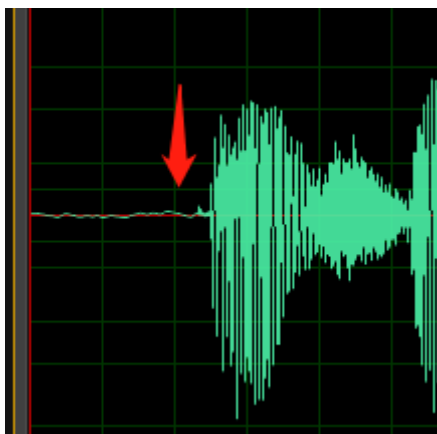


- The silent parts of the recording aren't clean; you can hear sounds such as ambient noise, mouth noise and echo.

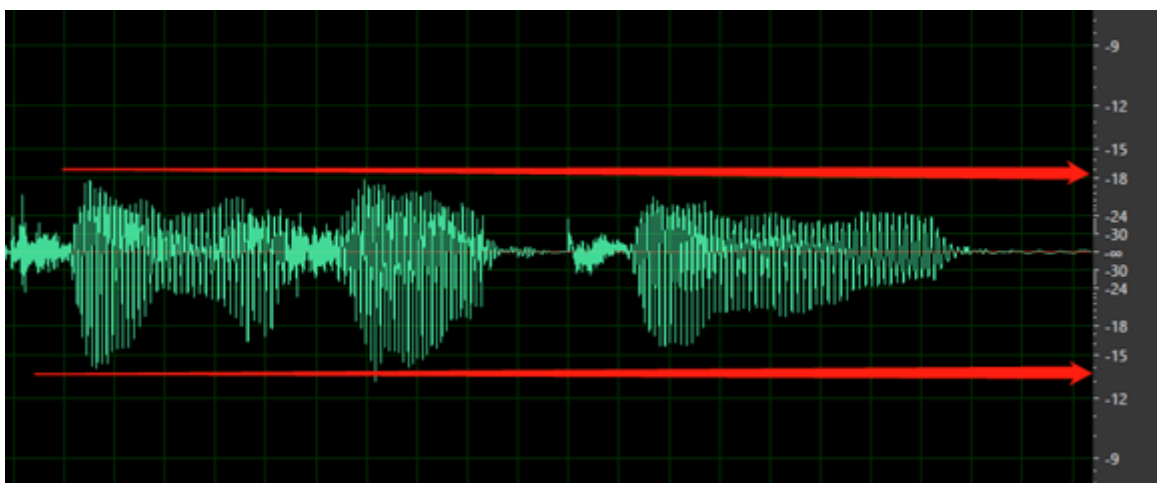
For example, below audio contains the environment noise between speeches.



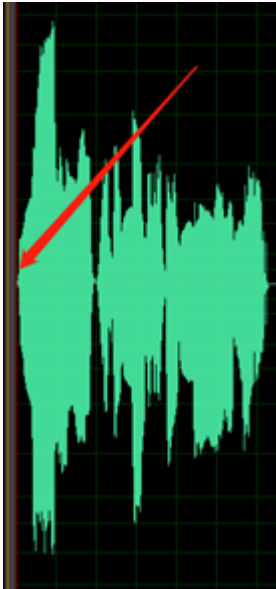
Below sample contains signs of DC offset or echo.



- The overall volume is too low. Your data is tagged as an issue if the volume is lower than -18 dB (10% of max volume). Make sure all audio files should be consistent at the same level of volume.



- No silence before the first word or after the last word. Also, the start or end silence shouldn't be longer than 200 ms or shorter than 100 ms.



Do it yourself

If you want to make the recording yourself, instead of going into a recording studio, here's a short primer. Thanks to the rise of home recording and podcasting, it's easier than ever to find good recording advice and resources online.

Your "recording booth" should be a small room with no noticeable echo or "room tone." It should be as quiet and soundproof as possible. Drapes on the walls can be used to reduce echo and neutralize or "deaden" the sound of the room.

Use a high-quality studio condenser microphone ("mic" for short) intended for recording voice. Sennheiser, AKG, and even newer Zoom mics can yield good results. You can buy a mic, or rent one from a local audio-visual rental firm. Look for one with a USB interface. This type of mic conveniently combines the microphone element, preamp, and analog-to-digital converter into one package, simplifying hookup.

You can also use an analog microphone. Many rental houses offer "vintage" microphones known for their voice character. Professional analog gear uses balanced XLR connectors, rather than the 1/4-inch plug that's used in consumer equipment. If you go analog, you'll also need a preamp and a computer audio interface with these connectors.

Install the microphone on a stand or boom, and install a pop filter in front of the microphone to eliminate noise from "plosive" consonants like "p" and "b." Some microphones come with a suspension mount that isolates them from vibrations in the stand, which is helpful.

The voice talent must stay at a consistent distance from the microphone. Use tape on the floor to mark where they should stand. If the talent prefers to sit, take special care to monitor mic

distance and avoid chair noise.

Use a stand to hold the script. Avoid angling the stand so that it can reflect sound toward the microphone.

The person operating the recording equipment — the recording engineer — should be in a separate room from the talent, with some way to talk to the talent in the recording booth (a *talkback circuit*).

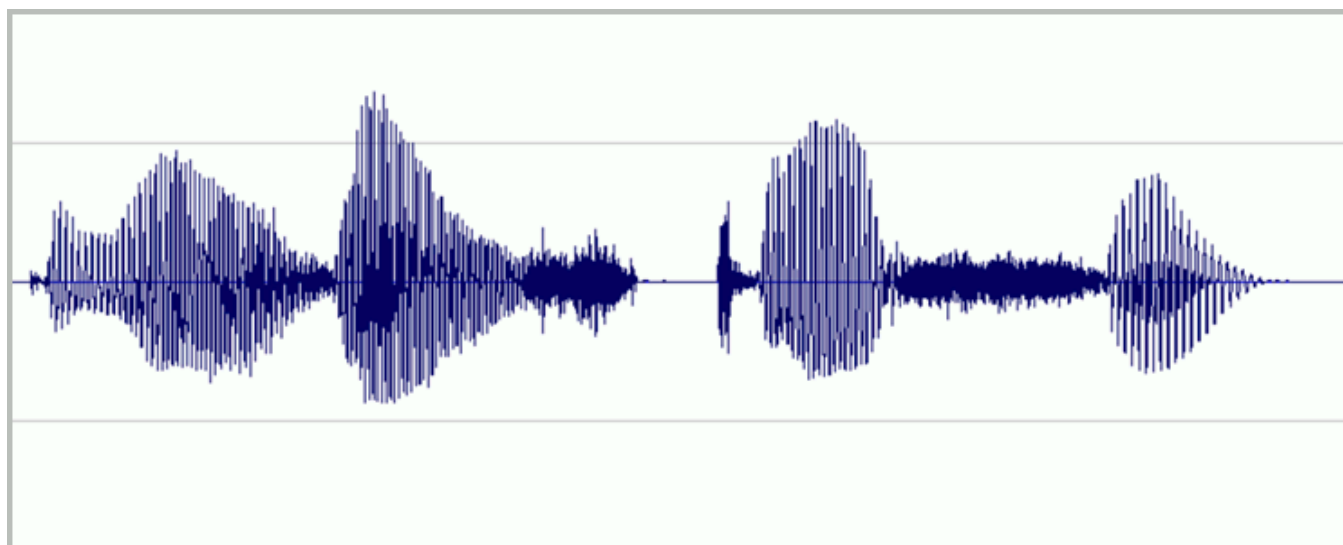
The recording should contain as little noise as possible, with a goal of -80 dB.

Listen closely to a recording of silence in your "booth," figure out where any noise is coming from, and eliminate the cause. Common sources of noise are air vents, fluorescent light ballasts, traffic on nearby roads, and equipment fans (even notebook PCs might have fans). Microphones and cables can pick up electrical noise from nearby AC wiring, usually a hum or buzz. A buzz can also be caused by a *ground loop*, which is caused by having equipment plugged into more than one electrical circuit.

💡 Tip

In some cases, you might be able to use an equalizer or a noise reduction software plug-in to help remove noise from your recordings, although it is always best to stop it at its source.

Set levels so that most of the available dynamic range of digital recording is used without overdriving. That means set the audio loud, but not so loud that it becomes distorted. An example of the waveform of a good recording is shown in the following image:



Here, most of the range (height) is used, but the highest peaks of the signal don't reach the top or bottom of the window. You can also see that the silence in the recording approximates a

thin horizontal line, indicating a low noise floor. This recording has acceptable dynamic range and signal-to-noise ratio.

Record directly into the computer via a high-quality audio interface or a USB port, depending on the mic you're using. For analog, keep the audio chain simple: mic, preamp, audio interface, computer. You can license both [Avid Pro Tools](#) and [Adobe Audition](#) monthly at a reasonable cost. If your budget is extremely tight, try the free [Audacity](#).

Record at 44.1 KHz 16 bit monophonic (CD quality) or better. Current state-of-the-art is 48 KHz 24 bit, if your equipment supports it. You'll down-sample your audio to 24 KHz 16-bit before you submit it to Speech Studio. Still, it pays to have a high-quality original recording in the event that edits are needed.

Ideally, have different people serve in the roles of director, engineer, and talent. Don't try to do it all yourself. In a pinch, one person can be both the director and the engineer.

Before the session

To avoid wasting studio time, run through the script with your voice talent before the recording session. While the voice talent becomes familiar with the text, they can clarify the pronunciation of any unfamiliar words.

 Note

Most recording studios offer electronic display of scripts in the recording booth. In this case, type your run-through notes directly into the script's document. You'll still want a paper copy to take notes on during the session, though. Most engineers will want a hard copy, too. And you'll still want a third printed copy as a backup for the talent in case the computer is down.

Your voice talent might ask which word you want emphasized in an utterance (the "operative word"). Tell them that you want a natural reading with no particular emphasis. Emphasis can be added when speech is synthesized; it shouldn't be a part of the original recording.

Direct the talent to pronounce words distinctly. Every word of the script should be pronounced as written. Sounds shouldn't be omitted or slurred together, as is common in casual speech, *unless they have been written that way in the script*.

Written text	Unwanted casual pronunciation
never going to give you up	never gonna give you up
there are four lights	there're four lights
how's the weather today	how's th' weather today
say hello to my little friend	say hello to my lil' friend

The talent shouldn't add distinct pauses between words. The sentence should still flow naturally, even while sounding a little formal. This fine distinction might take practice to get right.

The recording session

Create a reference recording, or *match file*, of a typical utterance at the beginning of the session. Ask the talent to repeat this line every page or so. Each time, compare the new recording to the reference. This practice helps the talent remain consistent in volume, tempo, pitch, and intonation. Meanwhile, the engineer can use the match file as a reference for levels and overall consistency of sound.

The match file is especially important when you resume recording after a break or on another day. Play it a few times for the talent and have them repeat it each time until they're matching well.

To record a corpus with a specific style, carefully choose scripts that showcase the desired style. During recording, ensure the voice talent maintains consistent in volume, tempo, pitch, and tone to achieve recordings that embody the intended style.

Coach your talent to take a deep breath and pause for a moment before each utterance. Record a couple of seconds of silence between utterances. Words should be pronounced the same way each time they appear, considering context. For example, "record" as a verb is pronounced differently from "record" as a noun.

Record approximately five seconds of silence before the first recording to capture the "room tone". This practice helps Speech Studio compensate for noise in the recordings.

Tip

All you need to capture is the voice talent, so you can make a monophonic (single-channel) recording of just their lines. However, if you record in stereo, you can use the second channel to record the chatter in the control room to capture discussion of

particular lines or takes. Remove this track from the version that's uploaded to Speech Studio.

Listen closely, using headphones, to the voice talent's performance. You're looking for good but natural diction, correct pronunciation, and a lack of unwanted sounds. Don't hesitate to ask your talent to re-record an utterance that doesn't meet these standards.

Tip

If you are using a large number of utterances, a single utterance might not have a noticeable effect on the resultant custom voice. It might be more expedient to simply note any utterances with issues, exclude them from your dataset, and see how your custom voice turns out. You can always go back to the studio and record the missed samples later.

Note the take number or time code on your script for each utterance. Ask the engineer to mark each utterance in the recording's metadata or cue sheet as well.

Take regular breaks and provide a beverage to help your voice talent keep their voice in good shape.

After the session

Modern recording studios run on computers. At the end of the session, you receive one or more audio files, not a tape. These files are probably WAV or AIFF format in CD quality (44.1 KHz 16-bit) or better. 24 KHz 16-bit is common and desirable. The default sampling rate for a custom voice is 24 KHz. It's recommended that you should use a sample rate of 24 KHz and higher for your training data. Higher sampling rates, such as 96 KHz, aren't usually needed.

Speech Studio requires each provided utterance to be in its own file. Each audio file delivered by the studio contains multiple utterances. So the primary post-production task is to split up the recordings and prepare them for submission. The recording engineer might have placed markers in the file (or provided a separate cue sheet) to indicate where each utterance starts.

Use your notes to find the exact takes you want, and then use a sound editing utility, such as [Avid Pro Tools](#), [Adobe Audition](#), or the free [Audacity](#), to copy each utterance into a new file.

Listen to each file carefully. At this stage, you can edit out small unwanted sounds that you missed during recording, like a slight lip smack before a line, but be careful not to remove any actual speech. If you can't fix a file, remove it from your dataset and note that you've done so.

Convert each file to 16 bits and a sample rate of 24 KHz and higher before saving and if you recorded the studio chatter, remove the second channel. Save each file in WAV format, naming the files with the utterance number from your script.

Finally, create the transcript that associates each WAV file with a text version of the corresponding utterance. [Train your voice model](#) includes details of the required format. You can copy the text directly from your script. Then create a Zip file of the WAV files and the text transcript.

Archive the original recordings in a safe place in case you need them later. Preserve your script and notes, too.

Next steps

You're ready to upload your recordings and create your custom voice.

[Train your voice model](#)

Create a project for professional voice

05/20/2025

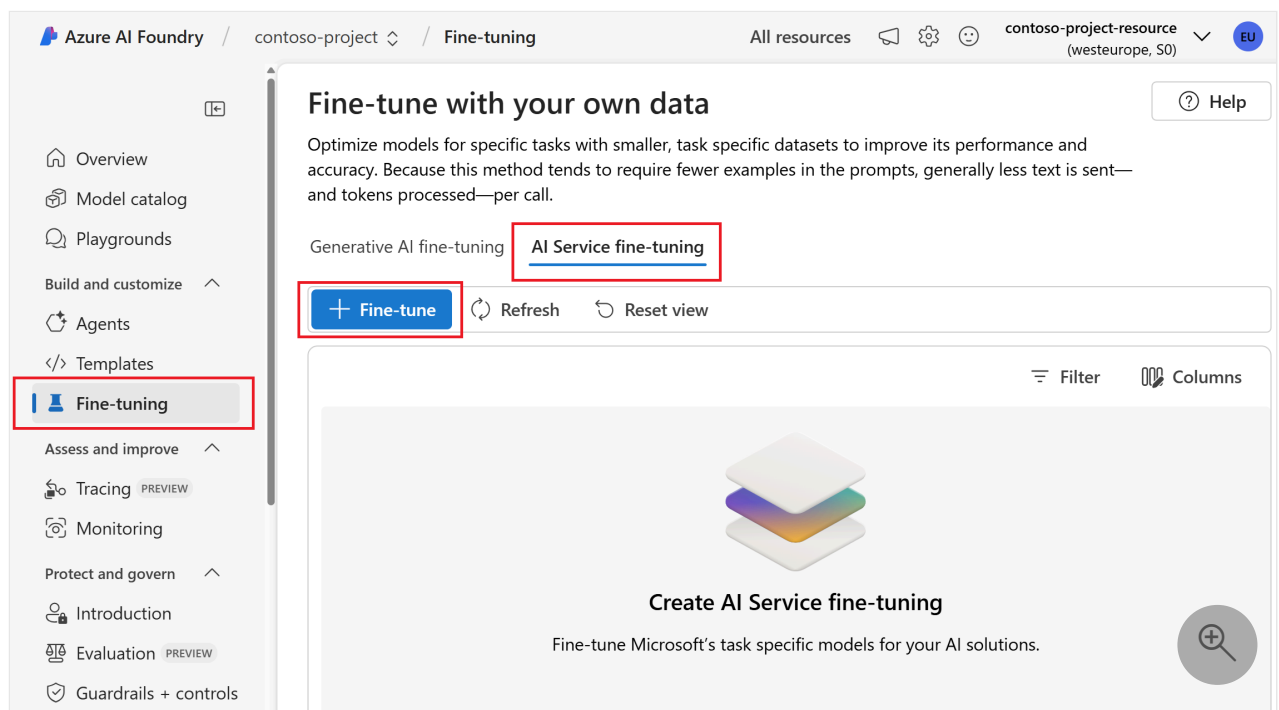
All it takes to get started are a handful of audio files and the associated transcriptions. See if custom voice supports your [language](#) and [region](#).

Start fine-tuning

In the [Azure AI Foundry portal](#), you can fine-tune some Azure AI services models. For example, you can fine-tune a professional voice model.

To fine-tune a professional voice model, follow these steps:

1. Go to your Azure AI Foundry project in the [Azure AI Foundry portal](#). If you need to create a project, see [Create an Azure AI Foundry project](#).
2. Select **Fine-tuning** from the left pane.
3. Select **AI Service fine-tuning** > **+ Fine-tune**.



4. In the wizard, select **Custom voice (professional voice fine-tuning)**.
5. Select **Next**.
6. Follow the instructions provided by the wizard to create your fine-tuning workspace.

Continue fine-tuning

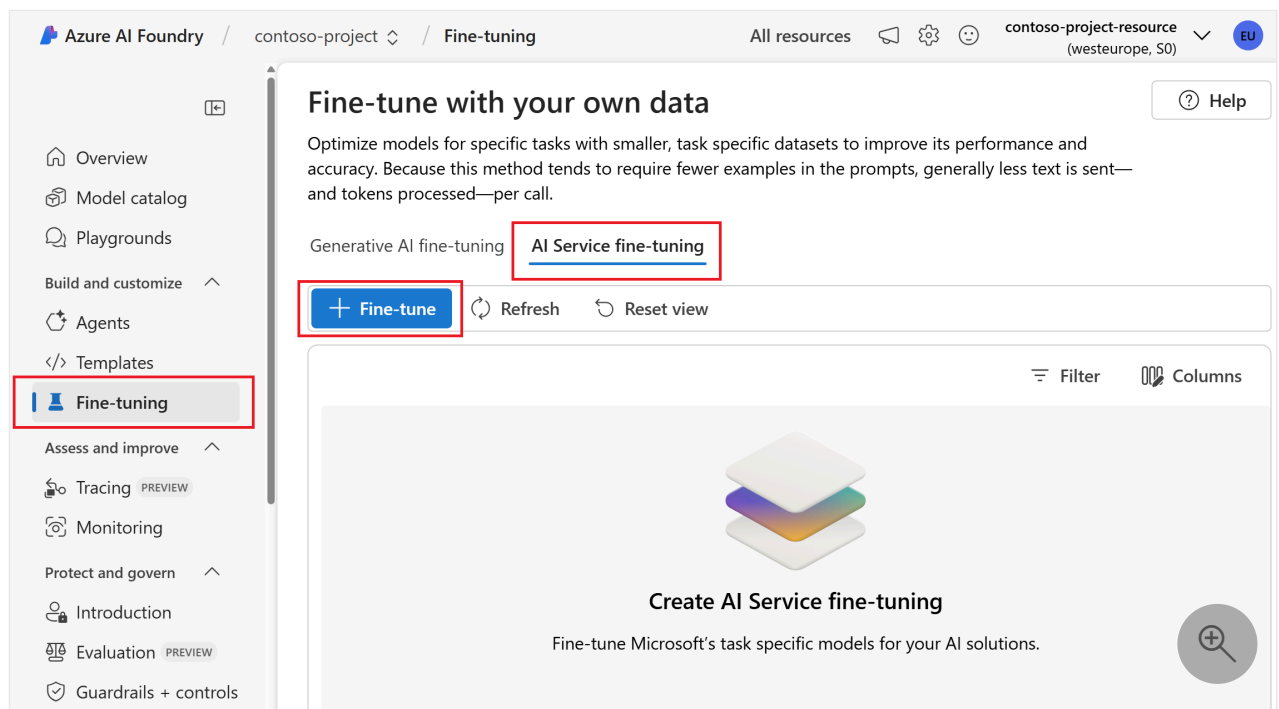
Go to the Azure AI Speech documentation to learn how to continue fine-tuning your professional voice model:

- [Add voice talent consent](#)
- [Add training datasets](#)
- [Train your voice model](#)
- [Deploy your professional voice model as an endpoint](#)

View fine-tuned models

After fine-tuning, you can access your custom voice models and deployments from the **Fine-tuning** page.

1. Sign in to the [Azure AI Foundry portal](#).
2. Select **Fine-tuning** from the left pane.
3. Select **AI Service fine-tuning**. You can view the status of your fine-tuning tasks and the models that were created.



Next steps

Add voice talent consent to the professional voice project.

Add voice talent consent to the professional voice project

05/20/2025

A voice talent is an individual or target speaker whose voices are recorded and used to create neural voice models.

Before you can fine-tune a professional voice, you must submit a recording of the voice talent's consent statement. The voice talent statement is a recording of the voice talent reading a statement that they consent to the usage of their speech data for professional voice fine-tuning. The consent statement is also used to verify that the voice talent is the same person as the speaker in the fine-tuning data.

Tip

Before you get started in Azure AI Foundry portal, define your voice [persona and choose the right voice talent](#).

You can find the verbal consent statement in multiple languages on [GitHub](#). The language of the verbal statement must be the same as your recording. See also the [disclosure for voice talent](#).

Add voice talent

Tip

For a sample consent statement and training data, see the [GitHub repository](#).

To add a voice talent profile and upload their consent statement, follow these steps:

1. Sign in to the [Azure AI Foundry portal](#).
2. Select **Fine-tuning** from the left pane and then select **AI Service fine-tuning**.
3. Select the professional voice fine-tuning task (by model name) that you [started as described in the create professional voice article](#).
4. Select **Set up voice talent** > **+ Add voice talent**.

5. In the **Add new voice talent** wizard, select the target scenarios for the voice talent. The target scenarios must be consistent with what you provided in the application form. The scenarios are used to help identify the voice talent and to ensure that the voice model is trained for the intended use cases.
6. Optionally in the **Voice characteristics** text box, enter a description of the characteristics of the voice you're going to create.
7. Select **Next**.
8. On the **Upload verbal statement** page, follow the instructions to upload the voice talent statement you recorded beforehand.
 - Enter the voice talent name and company name. The voice talent name must be the name of the person who recorded the consent statement. Enter the name in the same language used in the recorded statement. The company name must match the company name that was spoken in the recorded statement. Ensure the company name is entered in the same language as the recorded statement.
 - Make sure the verbal statement was [recorded](#) with the same settings, environment, and speaking style as your fine-tuning data.

The screenshot shows the 'Add new voice talent' wizard with three steps: 'Voice characteristics' (completed), 'Statement upload' (current), and 'Review and add'. The 'Statement upload' section contains instructions to upload an audio recording. Below the instructions are two text input fields: 'Voice talent name *' with the value 'Sample voice actor' and 'Company name *' with the value 'Contoso'. These fields are highlighted with a red rectangle. Below them is a text area for the verbal statement, which contains the text: "I **Sample voice actor** am aware that recordings of my voice will be used by **Contoso** to create and use a synthetic version of my voice." This text area is also highlighted with a red rectangle. At the bottom, there is a 'Verbal statement file *' label above a file upload area. Navigation buttons include 'Back', 'Next', 'Add voice talent', and 'Cancel'.

Add new voice talent

✓ Voice characteristics
● **Statement upload**
○ Review and add

Upload verbal statement

Upload an audio recording of your voice talent saying the sentence below. This audio file will be used to create a voice signature of your voice talent and to verify against your training data when you create a voice model. Learn more about [Microsoft's processing, use and retention of voice talent data](#).

Voice talent name *
Sample voice actor

Company name *
Contoso

"I **Sample voice actor** am aware that recordings of my voice will be used by **Contoso** to create and use a synthetic version of my voice."

Verbal statement file *

Back Next Add voice talent Cancel

9. Select **Next**.
10. Review the voice talent and persona details, and select **Add voice talent**.

After the voice talent status is *Succeeded*, you can [add fine-tuning data](#).

Next steps

Add training data for professional voice fine-tuning

Add a professional voice training dataset

05/20/2025

When you're ready to create a custom text to speech voice for your application, the first step is to gather audio recordings and associated scripts to start training the voice model. For details on recording voice samples, see [the tutorial](#). The Speech service uses this data to create a unique voice tuned to match the voice in the recordings. After you've trained the voice, you can start synthesizing speech in your applications.

All data you upload must meet the requirements for the data type that you choose. It's important to correctly format your data before it's uploaded, which ensures the data will be accurately processed by the Speech service. To confirm that your data is correctly formatted, see [Training data types](#).

ⓘ Note

- Standard subscription (S0) users can upload five data files simultaneously. If you reach the limit, wait until at least one of your data files finishes importing. Then try again.
- The maximum number of data files allowed to be imported per subscription is 500 .zip files for standard subscription (S0) users. Please see out [Speech service quotas and limits](#) for more details.

Upload your data

💡 Tip

For a sample consent statement and training data, see the [GitHub repository](#).[↗]

When you're ready to upload your data, go to the **Prepare training data** tab to add your first training set and upload data. A *training set* is a set of audio utterances and their mapping scripts used for training a voice model. You can use a training set to organize your training data. The service checks data readiness per each training set. You can import multiple data to a training set.

To upload training data, follow these steps:

1. Sign in to the [Azure AI Foundry portal](#).[↗]

2. Select **Fine-tuning** from the left pane and then select **AI Service fine-tuning**.
3. Select the professional voice fine-tuning task (by model name) that you [started as described in the create professional voice article](#).
4. Select **Prepare training data > Upload data**.
5. In the **Upload data** wizard, choose a [data type](#). If you're using the sample data, select **Individual utterances + matching transcript**.

Upload data [Close]

Data type (selected)

Training set selection

Data upload

Name and description

Review and upload

Choose a data type

Individual utterances + matching transcript

A collection (.zip) of audio files (.wav) as individual utterances, recommended to be 15 seconds or shorter, paired with a formatted transcript (.txt) denoting the audio file names.

Long audio + transcript [Preview]

A collection (.zip) of long, unsegmented audio files (.wav or .mp3), longer than 20 seconds, paired with a collection (.zip) of transcripts containing all of the spoken word.

Audio only [Preview]

A collection (.zip) of audio files (.wav or .mp3) without a transcript. A transcription for each spoken utterance will be automatically generated.

[Learn more about training data type](#)

Next [Upload data] [Cancel]

6. Select **Next**.
7. On the **Specify the target training set** page, select **Create new**.
8. Enter a training set name and then select **Create**.

9. Select **Next**.

10. On the **Data upload** page, select a **Recording file** and **Script file** in the respective tiles.
You can select local files from your computer or enter the Azure Blob storage URL to upload data.

11. Select **Next**.

12. Enter a name and description for your data and then select **Next**.

13. Review the upload details, and select **Upload data**.

ⓘ Note

Duplicate IDs aren't accepted. Utterances with the same ID will be removed.

Duplicate audio names are removed from the training. Make sure the data you select don't contain the same audio names within the .zip file or across multiple .zip files. If utterance IDs (either in audio or script files) are duplicates, they're rejected.

Data files are automatically validated when you select **Upload data**. Data validation includes series of checks on the audio files to verify their file format, size, and sampling rate. If there are any errors, fix them and submit again.

After you upload the data, you can check the details in the training set detail view. On the detail page, you can further check the pronunciation issue and the noise level for each of your data. The pronunciation score at the sentence level ranges from 0-100. A score below 70

normally indicates a speech error or script mismatch. Utterances with an overall score lower than 70 will be rejected. A heavy accent can reduce your pronunciation score and affect the generated digital voice.

Resolve data issues online

After upload, you can check the data details of the training set. Before continuing to [train your voice model](#), you should try to resolve any data issues.

Typical data issues

The issues are divided into three types. Refer to the following tables to check the respective types of errors.

Auto-rejected

Data with these errors won't be used for training. Imported data with errors will be ignored, so you don't need to delete them. You can [fix these data errors online](#) or upload the corrected data again for training.

 Expand table

Category	Name	Description
Script	Invalid separator	You must separate the utterance ID and the script content with a Tab character.
Script	Invalid script ID	The script line ID must be numeric.
Script	Duplicated script	Each line of the script content must be unique. The line is duplicated with {}.
Script	Script too long	The script must be less than 1,000 characters.
Script	No matching audio	The ID of each utterance (each line of the script file) must match the audio ID.
Script	No valid script	No valid script is found in this dataset. Fix the script lines that appear in the detailed issue list.
Audio	No matching script	No audio files match the script ID. The name of the .wav files must match with the IDs in the script file.
Audio	Invalid audio format	The audio format of the .wav files is invalid. Check the .wav file format by using an audio tool like SoX .
Audio	Low sampling rate	The sampling rate of the .wav files can't be lower than 16 KHz.

Category	Name	Description
Audio	Too long audio	Audio duration is longer than 30 seconds. Split the long audio into multiple files. It's a good idea to make utterances shorter than 15 seconds.
Audio	No valid audio	No valid audio is found in this dataset. Check your audio data and upload again.
Mismatch	Low scored utterance	Sentence-level pronunciation score is lower than 70. Review the script and the audio content to make sure they match.

Auto-fixed

The following errors are fixed automatically, but you should review and confirm the fixes are made correctly.

 Expand table

Category	Name	Description
Mismatch	Silence auto fixed	The start silence is detected to be shorter than 100 ms, and has been extended to 100 ms automatically. Download the normalized dataset and review it.
Mismatch	Silence auto fixed	The end silence is detected to be shorter than 100 ms, and has been extended to 100 ms automatically. Download the normalized dataset and review it.
Script	Text auto normalized	Text is automatically normalized for digits, symbols, and abbreviations. Review the script and audio to make sure they match.

Manual check required

Unresolved errors listed in the next table affect the quality of training, but data with these errors won't be excluded during training. For higher-quality training, it's a good idea to fix these errors manually.

 Expand table

Category	Name	Description
Script	Non-normalized text	This script contains symbols. Normalize the symbols to match the audio. For example, normalize <i>/</i> to <i>slash</i> .
Script	Not enough question utterances	At least 10 percent of the total utterances should be question sentences. This helps the voice model properly express a questioning tone.

Category	Name	Description
Script	Not enough exclamation utterances	At least 10 percent of the total utterances should be exclamation sentences. This helps the voice model properly express an excited tone.
Script	No valid end punctuation	Add one of the following at the end of the line: full stop (half-width '.' or full-width '。'), exclamation point (half-width '!' or full-width '！'), or question mark (half-width '?' or full-width '？').
Audio	Low sampling rate for neural voice	It's recommended that the sampling rate of your .wav files should be 24 KHz or higher for creating neural voices. If it's lower, it will be automatically raised to 24 KHz.
Volume	Overall volume too low	Volume shouldn't be lower than -18 dB (10 percent of max volume). Control the volume average level within proper range during the sample recording or data preparation.
Volume	Volume overflow	Overflowing volume is detected at {}s. Adjust the recording equipment to avoid the volume overflow at its peak value.
Volume	Start silence issue	The first 100 ms of silence isn't clean. Reduce the recording noise floor level, and leave the first 100 ms at the start silent.
Volume	End silence issue	The last 100 ms of silence isn't clean. Reduce the recording noise floor level, and leave the last 100 ms at the end silent.
Mismatch	Low scored words	Review the script and the audio content to make sure they match, and control the noise floor level. Reduce the length of long silence, or split the audio into multiple utterances if it's too long.
Mismatch	Start silence issue	Extra audio was heard before the first word. Review the script and the audio content to make sure they match, control the noise floor level, and make the first 100 ms silent.
Mismatch	End silence issue	Extra audio was heard after the last word. Review the script and the audio content to make sure they match, control the noise floor level, and make the last 100 ms silent.
Mismatch	Low signal-noise ratio	Audio SNR level is lower than 20 dB. At least 35 dB is recommended.

Category	Name	Description
Mismatch	No score available	Failed to recognize speech content in this audio. Check the audio and the script content to make sure the audio is valid, and matches the script.

Next steps

Train the professional voice

Train your professional voice model

05/20/2025

In this article, you learn how to fine-tune a professional voice through the Azure AI Foundry portal.

Important

Professional voice fine-tuning is currently only available in some regions. After your voice model is trained in a supported region, you can [copy the professional voice model](#) to an Azure AI Foundry resource in another region as needed. For more information, see the footnotes in the [Speech service table](#).

Training duration varies depending on how much data you use. It takes about 40 compute hours on average to fine-tune a professional voice. With an Azure AI Foundry standard (S0) resource, you can train four voices simultaneously. If you reach the limit, wait until at least one of your voice models finishes training, and then try again.

Note

Although the total number of hours required per [training method](#) varies, the same unit price applies to each. For more information, see the [custom neural training pricing details](#) .

Choose a training method

After you validate your data files, use them to build your custom voice model. When you create a custom voice, you can choose to train it with one of the following methods:

- **Neural:** Create a voice in the same language of your training data.
- **Neural - cross lingual:** Create a voice that speaks a different language from your training data. For example, with the `zh-CN` training data, you can create a voice that speaks `en-US`.

The language of the training data and the target language must both be one of the [languages that are supported](#) for cross lingual voice training. You don't need to prepare training data in the target language, but your test script must be in the target language.

- **Neural - multi style:** Create a custom voice that speaks in multiple styles and emotions, without adding new training data. Multiple style voices are useful for video game

characters, conversational chatbots, audiobooks, content readers, and more.

To create a multiple style voice, you need to prepare a set of general training data, at least 300 utterances. Select one or more of the preset target speaking styles. You can also create multiple custom styles by providing style samples, of at least 100 utterances per style, as extra training data for the same voice. The supported preset styles vary according to different languages. See [available preset styles across different languages](#).

- [Neural - multi lingual \(preview\)](#): Create a voice that speaks multiple languages using the single-language training data. For example, with the `en-US` primary training data, you can create a voice that speaks `en-US`, `de-DE`, `zh-CN` etc. secondary languages.

The primary language of the training data and the secondary languages must be in the [languages that are supported](#) for multi lingual voice training. You don't need to prepare training data in the secondary languages.

The language of the training data must be one of the [languages that are supported](#) for custom voice, cross-lingual, or multiple style training.

Train your custom voice model

To create a custom voice in Azure AI Foundry portal, follow these steps for one of the following methods:

Neural

1. Sign in to the [Azure AI Foundry portal](#) [↗](#).
2. Select **Fine-tuning** from the left pane and then select **AI Service fine-tuning**.
3. Select the professional voice fine-tuning task (by model name) that you [started as described in the create professional voice article](#).
4. Select **Train model** > **+ Train model**.
5. Select **Neural** as the [training method](#) for your model. To use a different training method, see [Neural - cross lingual](#), [Neural - multi style](#), [Neural - multi lingual \(preview\)](#), or [Neural - HD Voice \(preview\)](#).

Train a new model

Training method

- Training method (selected)
- Data and voice talent
- Test script
- Basic information
- Review and train

Select the training method for your model

Neural - Default
 Create a voice in the same language as your training data.

Neural - Cross lingual
 Create a voice that speaks a different language from your training data.

Neural - Multi lingual Preview
 Create a voice that speaks multiple languages using training data from a single language.

Neural - Multi style
 Create a voice that speaks in multiple styles.

Version * ⓘ
 V10.1

Version description: Model updated to understand context and automatically adjust speaking tones with contextual training data. 15 compute hours estimated for each training. Start with 300 utterances or 30 minutes of speech data to

Next **Train** **Cancel**

6. Select a version of the training recipe for your model. The latest version is selected by default. The supported features and training time can vary by version. Normally, we recommend the latest version. In some cases, you can choose an earlier version to reduce training time. See [Bilingual training](#) for more information about bilingual training and differences between locales.

7. Select **Next**.

8. Select the data that you want to use for training. Duplicate audio names are removed from the training. Make sure that the data you select doesn't contain the same audio names across multiple .zip files.

You can select only successfully processed datasets for training. If you don't see your training set in the list, check your data processing status.

9. Select a speaker file with the voice talent statement that corresponds to the speaker in your training data.

10. Select **Next**.

11. Select a test script and then select **Next**.

- Each training generates 100 sample audio files automatically to help you test the model with a default script.
- Alternatively, you can select **Add my own test script** and provide your own test script with up to 100 utterances to test the model at no extra cost. The generated audio files are a combination of the automatic test scripts and custom test scripts. For more information, see [test script requirements](#).

12. Enter a **Voice model name**. Choose a name carefully. The model name is used as the voice name in your [speech synthesis request](#) by the SDK and SSML input. Only letters, numbers, and a few punctuation characters are allowed. Use different names for different neural voice models.
13. Optionally, enter the **Description** to help you identify the model. A common use of the description is to record the names of the data that you used to create the model.
14. Select the checkbox to accept the terms of use and then select **Next**.
15. Review the settings and select the box to accept the terms of use.
16. Select **Train** to start training the model.

Bilingual training

If you select the **Neural** training type, you can train a voice to speak in multiple languages. The `zh-CN`, `zh-HK`, and `zh-TW` locales support bilingual training for the voice to speak both Chinese and English. Depending in part on your training data, the synthesized voice can speak English with an English native accent or English with the same accent as the training data.

ⓘ Note

To enable a voice in the `zh-CN` locale to speak English with the same accent as the sample data, you should upload English data to a **Contextual** training set, or choose `Chinese (Mandarin, Simplified)`, `English bilingual` when creating a project or specify the `zh-CN (English bilingual)` locale for the training set data via REST API.

In your contextual training set, include at least 100 sentences or 10 minutes of English content and do not exceed the amount of Chinese content.

The following table shows the differences among the locales:

[🔗 Expand table](#)

Speech Studio locale	REST API locale	Bilingual support
<code>Chinese (Mandarin, Simplified)</code>	<code>zh-CN</code>	If your sample data includes English, the synthesized voice speaks English with an English native accent, instead of the same accent as the sample data, regardless of the amount of English data.

Speech Studio locale	REST API locale	Bilingual support
Chinese (Mandarin, Simplified), English bilingual	zh-CN (English bilingual)	If you want the synthesized voice to speak English with the same accent as the sample data, we recommend including over 10% English data in your training set. Otherwise, the English speaking accent might not be ideal.
Chinese (Cantonese, Simplified)	zh-HK	If you want to train a synthesized voice capable of speaking English with the same accent as your sample data, make sure to provide over 10% English data in your training set. Otherwise, it defaults to an English native accent. The 10% threshold is calculated based on the data accepted after successful uploading, not the data before uploading. If some uploaded English data is rejected due to defects and doesn't meet the 10% threshold, the synthesized voice defaults to an English native accent.
Chinese (Taiwanese Mandarin, Traditional)	zh-TW	If you want to train a synthesized voice capable of speaking English with the same accent as your sample data, make sure to provide over 10% English data in your training set. Otherwise, it defaults to an English native accent. The 10% threshold is calculated based on the data accepted after successful uploading, not the data before uploading. If some uploaded English data is rejected due to defects and doesn't meet the 10% threshold, the synthesized voice defaults to an English native accent.

Monitor the training process

The **Train model** table displays a new entry that corresponds to this newly created model. The status reflects the process of converting your data to a voice model, as described in this table:

 Expand table

State	Meaning
Processing	Your voice model is being created.
Succeeded	Your voice model has been created and can be deployed.
Failed	Your voice model has failed in training. The cause of the failure might be, for example, unseen data problems or network issues.
Canceled	The training for your voice model was canceled.

While the model status is **Processing**, you can select the model and then select **Cancel training** to cancel training. You're not charged for this canceled training.

Train and review your voice model

Train a new model Deploy model Cancel training Edit Delete Copy to Clone model

Name	Description	Model type	Model language	Status	Engine version	Data size	Training
Contoso fine-tuned voice		Neural V10.1	English (United States)	Processing	2025.02.07.0	300	

After you finish training the model successfully, you can review the model details and [Test your voice model](#).

Rename your model

You have to clone your model to rename it. You can't rename the model directly.

1. Select the model.
2. Select **Clone model** to create a clone of the model with a new name in the current project.
3. Enter the new name on the **Clone voice model** window.
4. Select **Submit**. The text *Neural* is automatically added as a suffix to your new model name.

Test your voice model

After your voice model is successfully built, you can use the generated sample audio files to test it before you deploy it.

ⓘ Note

Neural - multi lingual (preview) and Neural - HD Voice (preview) do not support this type of testing.

The quality of the voice depends on many factors, such as:

- The size of the training data.
- The quality of the recording.
- The accuracy of the transcript file.

- How well the recorded voice in the training data matches the personality of the designed voice for your intended use case.

Select **DefaultTests** under **Testing** to listen to the sample audio files. The default test samples include 100 sample audio files generated automatically during training to help you test the model. In addition to these 100 audio files provided by default, your own test script utterances are also added to **DefaultTests** set. This addition is at most 100 utterances. You're not charged for the testing with **DefaultTests**.

If you want to upload your own test scripts to further test your model, select **Add test scripts** to upload your own test script.

Before you upload test script, check the [Test script requirements](#). You're charged for the extra testing with the batch synthesis based on the number of billable characters. See [Azure AI Speech pricing](#).

Under **Add test scripts**, select **Browse for a file** to select your own script, then select **Add** to upload it.

Test script requirements

The test script must be a `.txt` file that is less than 1 MB. Supported encoding formats include ANSI/ASCII, UTF-8, UTF-8-BOM, UTF-16-LE, or UTF-16-BE.

Unlike the [training transcription files](#), the test script should exclude the utterance ID, which is the filename of each utterance. Otherwise, these IDs are spoken.

Here's an example set of utterances in one `.txt` file:

text

```
This is the waistline, and it's falling.  
We have trouble scoring.  
It was Janet Maslin.
```

Each paragraph of the utterance results in a separate audio. If you want to combine all sentences into one audio, make them a single paragraph.

ⓘ Note

The generated audio files are a combination of the automatic test scripts and custom test scripts.

Update engine version for your voice model

Azure text to speech engines are updated from time to time to capture the latest language model that defines the pronunciation of the language. After you train your voice, you can apply your voice to the new language model by updating to the latest engine version.

- When a new engine is available, you're prompted to update your neural voice model.
- Go to the model details page and follow the on-screen instructions to install the latest engine.
- Alternatively, select **Install the latest engine** later to update your model to the latest engine version. You're not charged for engine update. The previous versions are still kept.
- You can check all engine versions for the model from the **Engine version** list, or remove one if you don't need it anymore.

The updated version is automatically set as default. But you can change the default version by selecting a version from the drop-down list and selecting **Set as default**.

If you want to test each engine version of your voice model, you can select a version from the list, then select **DefaultTests** under **Testing** to listen to the sample audio files. If you want to upload your own test scripts to further test your current engine version, first make sure the version is set as default, then follow the steps in [Test your voice model](#).

Updating the engine creates a new version of the model at no extra cost. After you update the engine version for your voice model, you need to deploy the new version to [create a new endpoint](#). You can only deploy the default version.

After you create a new endpoint, you need to [transfer the traffic to the new endpoint in your product](#).

To learn more about the capabilities and limits of this feature, and the best practice to improve your model quality, see [Characteristics and limitations for using custom voice](#).

Copy your voice model to another project

ⓘ Note

In this context "project" refers to a fine-tuning task rather than an Azure AI Foundry project.

After training you can copy your voice model to another project for the same region or another region.

For example, you can copy a professional voice model that was trained in one region, to a project for another region. Professional voice fine-tuning is currently only [available in some regions](#).

To copy your custom voice model to another project:

1. On the **Train model** tab, select a voice model that you want to copy, and then select **Copy to project**.
2. Select the **Subscription**, **Target region**, **Connected AI Service resource** (AI Foundry resource), and **Target fine-tuning task** where you want to copy the model.
3. Select **Copy to** to copy the model.
4. Select **View model** under the notification message for the successful copying.

Navigate to the project where you copied the model to [deploy the model copy](#).

Next steps

Deploy the professional voice endpoint

Deploy your professional voice model as an endpoint

05/20/2025

After you successfully created and [fine-tuned](#) your professional voice model, you deploy it to a custom voice endpoint.

ⓘ Note

You can create up to 50 endpoints with a standard (S0) Speech resource, each with its own custom voice.

To use your fine-tuned professional voice, you must specify the voice model name, use the custom URI directly in an HTTP request, and use the same Speech resource to pass through the authentication of the text to speech service.

Add a deployment endpoint

To create a professional voice endpoint:

To deploy an endpoint, follow these steps:

1. Sign in to the [Azure AI Foundry portal](#).
2. Select **Fine-tuning** from the left pane and then select **AI Service fine-tuning**.
3. Select the professional voice fine-tuning task (by model name) that you [started as described in the create professional voice article](#).
4. Select **Deploy model** > **Deploy model**.
5. Select a voice model that you want to associate with this endpoint and then select **Next**.
6. Enter a **Endpoint name** and **Description** for your custom endpoint.
7. Select **Endpoint type** according to your scenario. If your resource is in a supported region, the default setting for the endpoint type is *High performance*. Otherwise, if the resource is in an unsupported region, the only available option is *Fast resume*.
 - **High performance:** Optimized for scenarios with real-time and high-volume synthesis requests, such as conversational AI, call-center bots. It takes around 5 minutes to deploy or resume an endpoint. For information about regions where the *High performance* endpoint type is supported, see the footnotes in the [regions](#) table.
 - **Fast-resume:** Optimized for audio content creation scenarios with less frequent synthesis requests. Easy and quick to deploy or resume an endpoint in under a

minute. The fast-resume endpoint type is supported in all [regions](#) where text to speech is available.

8. Select **Next**.
9. Select the checkbox to accept the terms of use and then select **Next**.
10. Review the settings and select the box to accept the model hosting costs.
11. Select **Deploy** to create your endpoint.

After your endpoint is deployed, the endpoint name appears as a link. Select the link to display information specific to your endpoint, such as the endpoint key, endpoint URL, and sample code. When the status of the deployment is **Succeeded**, the endpoint is ready for use.

Use your custom voice

The custom endpoint is functionally identical to the standard endpoint that's used for text to speech requests.

One difference is that the `EndpointId` must be specified to use the custom voice via the Speech SDK. You can start with the [text to speech quickstart](#) and then update the code with the `EndpointId` and `SpeechSynthesisVoiceName`. For more information, see [use a custom endpoint](#).

To use a custom voice via [Speech Synthesis Markup Language \(SSML\)](#), specify the model name as the voice name. This example uses the `YourCustomVoiceName` voice.

XML

```
<speak version="1.0" xmlns="http://www.w3.org/2001/10/synthesis" xml:lang="en-US">
  <voice name="YourCustomVoiceName">
    This is the text that is spoken.
  </voice>
</speak>
```

Switch to a new voice model in your product

Once you updated your voice model to the latest engine version, or if you want to switch to a new voice in your product, you need to redeploy the new voice model to a new endpoint. Redeploying new voice model on your existing endpoint isn't supported. After deployment, switch the traffic to the newly created endpoint. We recommend that you transfer the traffic to the new endpoint in a test environment first to ensure that the traffic works well, and then transfer to the new endpoint in the production environment. During the transition, you need to keep the old endpoint. If there are some problems with the new endpoint during transition, you can switch back to your old endpoint. If the traffic has been running well on the new endpoint for about 24 hours (recommended value), you can delete your old endpoint.

ⓘ Note

If your voice name is changed and you're using Speech Synthesis Markup Language (SSML), be sure to use the new voice name in SSML.

Suspend and resume an endpoint

You can suspend or resume an endpoint to limit spend and conserve resources that aren't in use. You aren't charged while the endpoint is suspended. When you resume an endpoint, you can continue to use the same endpoint URL in your application to synthesize speech.

ⓘ Note

The suspend operation completes almost immediately. The resume operation completes in about the same amount of time as a new deployment.

This section describes how to suspend or resume a custom voice endpoint in the Azure AI Foundry portal.

Suspend endpoint

To suspend and deactivate your endpoint:

1. Sign in to the [Azure AI Foundry portal](#).
2. Select **Fine-tuning** from the left pane and then select **AI Service fine-tuning**.
3. Select the professional voice fine-tuning task (by model name) that you [started as described in the create professional voice article](#).
4. Select **Deploy model**.
5. Select the endpoint you want to suspend and then select **Suspend**.

Contoso custom voice

Professional voice | English (United States)

Male

- Getting started
- Set up voice talent
- Prepare training data
- Train model
- Deploy model**

Deploy your model

Deploy model | **Suspend** | Resume | Edit | Delete

Name	Description	Voice name	Model...	Model langua...	Created on	Status
Contoso fine-tuned voice		Contoso fine-tuned voice	Neural V10.1	English (United States)	5/15/2025, 7:15 AM	Succeeded

6. In the dialog box that appears, select **Suspend**. After the endpoint is suspended, the status changes from **Succeeded** to **Suspended**.

Resume endpoint

1. To resume and activate your endpoint, select **Resume** from the **Deploy model** tab in the [Azure AI Foundry portal](#) .
2. In the dialog box that appears, select **Submit**. After you successfully reactivate the endpoint, the status will change from **Suspended** to **Succeeded**.

Next steps

- Learn more about custom voice in the [overview](#).
- Learn more about custom avatar in the [overview](#).

Custom voice lite

06/02/2025

Custom voice lite is a project type where you can demo and evaluate custom voice before investing in professional recordings to create a higher-quality voice. No application is required for demo and evaluation purposes. However, Microsoft restricts and selects the recording and testing samples for use with custom voice lite. You must apply for full access to professional voice fine-tuning in order to deploy and use the custom voice lite model for business purpose. In that case, request access on the [intake form](#).

 Note

Custom voice lite is only available in the [Speech Studio](#). It isn't available through the Azure AI Foundry portal, REST API, or SDKs.

With a custom voice lite project, you record your voice online by reading 20-50 pre-defined scripts provided by Microsoft. After you've recorded at least 20 samples, you can start to train a model. Once the model is trained successfully, you can review the model and check out 20 output samples produced with another set of pre-defined scripts.

See the [supported languages](#) for custom voice.

Compare project types

The following table summarizes key differences between custom voice lite and professional voice fine-tuning.

 **Expand table**

Items	Lite	Professional
Target scenarios	Demonstration or evaluation	Professional scenarios like brand and character voices for chat bots, or audio content reading.
Training data	Record online using Speech Studio	Bring your own data. Recording in a professional studio is recommended.
Scripts for recording	Provided in Speech Studio	Use your own scripts that match the use case scenario. Microsoft provides example scripts for reference.

Items	Lite	Professional
Required data size	20-50 utterances	300-2000 utterances
Training time	Less than one compute hour	Approximately 20-40 compute hours
Voice quality	Moderate quality	High quality
Availability	Anyone can record samples online and train a model for demo and evaluation purpose. Full access to custom voice is required if you want to deploy the custom voice lite model for business use.	Data upload isn't restricted, but you can only fine-tune a professional voice after access is approved. Professional voice fine-tuning is limited based on eligibility and usage criteria. Request access on the intake form .
Pricing	Per unit prices apply equally for both custom voice lite and professional voice fine-tuning. Check the pricing details here .	Per unit prices apply equally for both custom voice lite and professional voice fine-tuning. Check the pricing details here .

Create a custom voice lite project

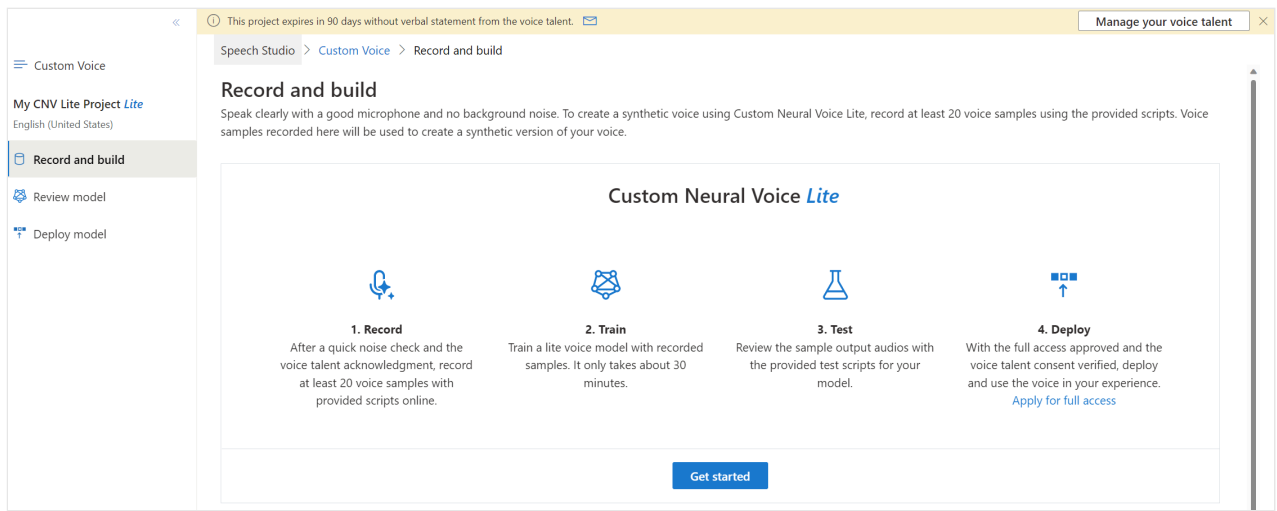
To create a custom voice lite project, follow these steps:

1. Sign in to the [Speech Studio](#).
2. Select the subscription and Speech resource to work with.
3. Select **Custom voice** > **Create a project**.
4. Select **Custom neural voice lite** > **Next**. To create a custom voice professional project instead, see the [professional voice fine-tuning documentation](#).
5. Follow the instructions provided by the wizard to create your project.

Important

The custom voice lite project expires after 90 days unless the [verbal statement](#) recorded by the voice talent is submitted.

6. Select the new project by name or select **Go to project**. You see these menu items in the left panel: **Record and build**, **Review model**, and **Deploy model**.



Record and build a custom voice lite model

Record at least 20 voice samples (up to 50) with provided scripts online. Voice samples recorded here are used to create a synthetic version of your voice.

ⓘ Note

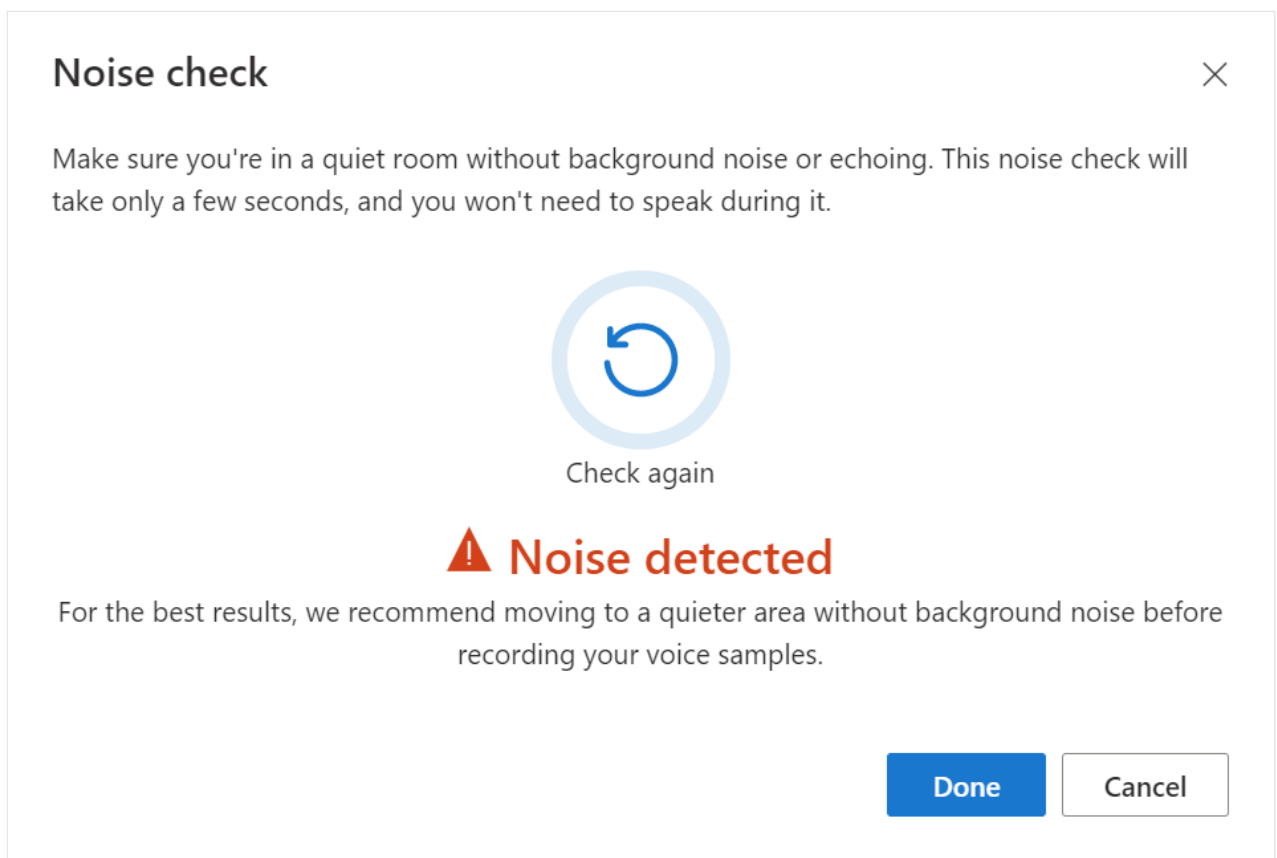
Custom voice training is currently only available in some regions. See footnotes in the [regions](#) table for more information.

Here are some tips to help you record your voice samples:

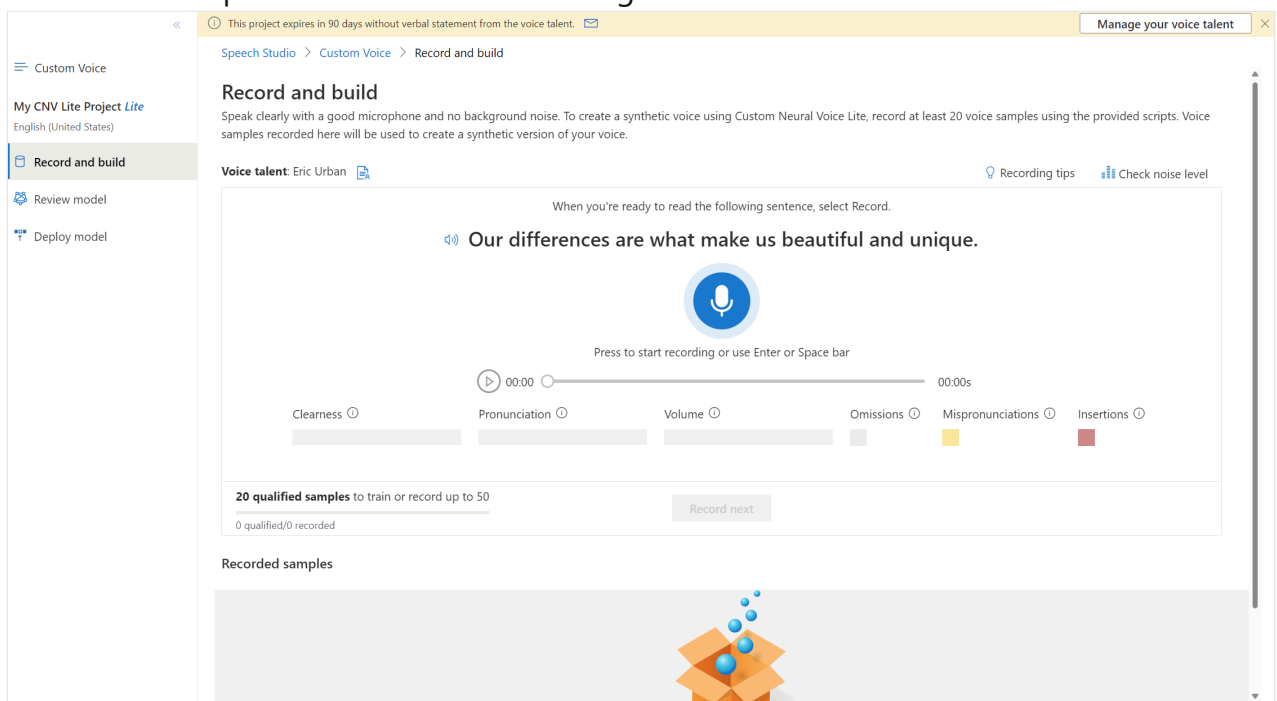
- Use a good microphone. Increase the clarity of your samples by using a high-quality microphone. Speak about 8 inches away from the microphone to avoid mouth noises.
- Avoid background noise. Record in a quiet room without background noise or echoing.
- Relax and speak naturally. Allow yourself to express emotions as you read the sentences.
- Record in one take. To keep a consistent energy level, record all sentences in one session.
- Pronounce each word correctly, and speak clearly.

To record and build a custom voice lite model, follow these steps in [Speech Studio](#):

1. Select **Custom voice** > Your project name > **Record and build**.
2. Select **Get started**.
3. Read the Voice talent terms of use carefully. Select the checkbox to acknowledge the terms of use.
4. Select **Accept**
5. Press the microphone icon to start the noise check. This noise check takes only a few seconds, and you don't need to speak during it.
6. If noise was detected, you can select **Check again** to repeat the noise check. If no noise was detected, you can select **Done** to proceed to the next step.



7. Review the recording tips and select **Got it**. For the best results, go to a quiet area without background noise before recording your voice samples.
8. Press the microphone icon to start recording.



9. Press the stop icon to stop recording.
10. Review quality metrics. After recording each sample, check its quality metric before continuing to the next one.
11. Record more samples. Although you can create a model with just 20 samples, it's recommended that you record up to 50 to get better quality.
12. Select **Train model** to start the training process.

The training process takes approximately one compute hour. You can check the progress of the training process in the **Review model** page.

Review model

To review the custom voice lite model and listen to your own synthetic voice, follow these steps:

1. Select **Custom voice** > Your project name > **Review model**. Here you can review the voice model name, model language, sample data size, and training progress. The voice name is composed of the word "Neural" appended to your project name.
2. Select the voice model name to review the model details and listen to the sample text to speech results.
3. Select the play icon to hear your voice speak each script.

Custom Voice

My CNV Lite Project *Lite*
English (United States)

Record and build

Review model

Deploy model

This project expires in 90 days without full access approval ([Learn more](#)) to Custom Neural Voice. [Apply for access](#)

Speech Studio > Custom Voice > Review model > My CNV Lite ProjectNeural

My CNV Lite ProjectNeural

Gender	Model language	Model type	Utterances ⓘ	Sampling rate	Status	Model ID	Created
Male	English (United States)	Neural	20	24000	⊙ Succeeded	ae6db871-18cd-42ab-856f-90282d9c5fb3	10/15/2022 8:18 AM

Sample output

Delete

Text input	Audio
✓ Congratulations! You have successfully created your synthetic voice with Custom Neural Voice Lite!	▶ ○
To deploy your voice model and use it in your business applications, you must get the full access to Custom Neural Voice and the explicit consent from your voice talent.	▶ ○
With the full access, you can create an even more natural voice by bringing your own data recorded in professional studios, using Custom Neural Voice Pro.	▶ ○

Submit verbal statement

A verbal statement recorded by the voice talent is required before you can [deploy the model](#) for your business use.

To submit the voice talent verbal statement, follow these steps in [Speech Studio](#) ↗:

1. Select **Custom voice** > Your project name > **Deploy model** > **Manage your voice talent**.

The screenshot shows the 'Manage your voice talent' page in the Speech Studio interface. The left sidebar contains a navigation menu with 'Custom Voice' selected, and a sub-menu with 'My CNV Lite Project Lite' (English (United States)), 'Record and build', 'Review model', and 'Deploy model'. The main content area has a breadcrumb trail: 'Speech Studio > Custom Voice > Voice talent statement'. Below this, there are two tabs: 'Deploy your model' and 'Manage your voice talent' (which is active). A warning banner at the top states: 'This project expires in 90 days without full access approval (Learn more) to Custom Neural Voice.' Below the tabs, there is a text block: 'Before you deploy the model, make sure your get meaningful consent from the voice talent to use the voice in your apps for the specific scenario defined. Learn more about Microsoft's processing, use and retention of voice talent data.' The form contains three input fields: 'Model *' (a dropdown menu showing 'Select model'), 'Voice talent name *' (a text box with placeholder '[state your first and last name]'), and 'Company name *' (a text box with placeholder '[state the name of the company]'). Below these fields is a large text area with the following text: 'When you're ready to read the following sentence, select Record. I, [your first and last name], understand that the synthetic voice designated to sound like my voice created with Microsoft Speech technology will be used by [the name of the company] for its own business purposes.' Below the text area is a microphone icon with a blue circle around it. Below the microphone icon is a progress bar with a play button icon on the left, a slider, and a '00:00s' label on the right. Below the progress bar are six checkboxes: 'Clearness', 'Pronunciation', 'Volume', 'Omissions', 'Mispronunciations', and 'Insertions'. Below these checkboxes is a 'Submit' button.

2. Select the model.
3. Enter the voice talent name and company name.
4. Read and record the statement. Select the microphone icon to start recording. Select the stop icon to stop recording.
5. Select **Submit** to submit the statement.
6. Check the processing status in the script table at the bottom of the dashboard. Once the status is **Succeeded**, you can [deploy the model](#).

Deploy model

To deploy your custom voice lite model and use it in your applications, you must get the full access to custom voice. Request access on the [intake form](#). Within approximately 10 business days, you receive an email with the approval status. A [verbal statement](#) recorded by the voice talent is also required before you can deploy the model for your business use.

To deploy a custom voice lite model, follow these steps in [Speech Studio](#):

1. Select **Custom voice** > Your project name > **Deploy model** > **Deploy model**.
2. Select a voice model name and then select **Next**.
3. Enter a name and description for your endpoint and then select **Next**.
4. Select the checkbox to agree to the terms of use and then select **Next**.
5. Select **Deploy** to deploy the model.

From here, you can use the custom voice lite model similarly as you would use a professional voice model. For example, you can [suspend or resume](#) an endpoint after it's created, to limit spend and conserve resources that aren't in use. You can also access the voice in the [Audio Content Creation](#) tool in the [Speech Studio](#).

Next steps

- [Fine-tune a professional voice](#)
- [Try the text to speech quickstart](#)
- [Learn more about speech synthesis](#)

What is personal voice for text to speech?

With personal voice, you can enable your users to get AI generated replication of their own voices in a few seconds. With a verbal statement and a short speech sample as the audio prompt, you can create a personal voice for your users and allow them to generate speech in any of the more than 90 languages supported across more than 100 locales.

ⓘ Note

For the current list of regions that support personal voice, see the [Speech service regions table](#). For supported locales, see [personal voice language support](#).

The following table summarizes the difference between personal voice and professional voice.

 Expand table

Comparison	Personal voice	Professional voice
Target scenarios	Business customers to build an app to allow their users to create and use their own personal voice in the app.	Professional scenarios like brand and character voices for chat bots, or audio content reading.
Use cases	Restricted to limited use cases. See the transparency note .	
Training data	Make sure you follow the code of conduct.	Bring your own data. Recording in a professional studio is recommended.
Required data size	One minute of human speech.	300-2000 utterances (about 30 minutes to 3 hours of human speech).
Training time	Less than 5 seconds	Approximately 20-40 compute hours.
Voice quality	Natural	Highly natural
Multilingual support	Yes. The voice is able to speak about 100 languages, with automatic language detection enabled.	Yes. You need to select the "Neural – cross lingual" feature to train a model that speaks a different language from the training data.
Availability	The demo on Speech Studio  is available upon registration. Access to the API is restricted to eligible customers and approved use cases. Request access through the intake form.	You can only use professional voice fine-tuning after access is approved. Professional voice fine-tuning access is limited based on eligibility and usage criteria. Request access through the intake form.
Pricing	Check the pricing details here  ¹ .	Check the pricing details here  .

Comparison	Personal voice	Professional voice
Responsible AI requirements	Speaker's verbal statement required. No unapproved use case allowed.	Speaker's verbal statement required. No unapproved use case allowed.

¹ Note that personal voice pricing will only be visible for service regions where the feature is available. For the current list of supported regions, see the [Speech service regions table](#).

Try the demo

If you have an S0 resource, you can access the personal voice demo in Speech Studio. To use the personal voice API, you can apply for access [here](#).

1. Go to [Speech Studio](#).
2. Select the **Personal Voice** card.
3. You can record your own voice and try the voice output samples in different languages. The demo includes a subset of the languages supported by personal voice.

Try your own voice

Record your voice and create an AI replica in seconds. With each standard Speech resource, you can try up to five voices for free. If you want to use this service for your business, please apply for access through [this link](#).

Use with your resource

☒ I acknowledge that this application uses the resource **tianhua-speech-s0-eus** and will incur usage to my account. [Choose a different resource](#)

Voice list

[+ New voice](#)

✓ Jessa Smith_20231110_161816

Detail

Jessa Smith_20231110_161816

Voice talent	Verbal statement	Source language
Jessa Smith	▶	English (United States)

Generate audios using sample scripts

Output language English (United States) ▼

The two reservoirs fed by the Colorado River watershed, provide a critical supply of drinking water an...	▶	↓
Let me give you the container number here.	▶	↓
I created a ticket number for you as a reference to this call.	▶	↓
I've never been there.	▶	↓
The monkey was confident since he was good at imitating.	▶	↓

4/5 Voice

How to create a personal voice

To get started, here's a summary of the steps to create a personal voice:

1. [Create a project](#).
2. [Upload consent file](#). With the personal voice feature, it's required that every voice be created with explicit consent from the user. A recorded statement from the user is required acknowledging that the customer (Azure AI Speech resource owner) will create and use their voice.
3. [Get a speaker profile ID](#) for the personal voice. You get a speaker profile ID based on the speaker's verbal consent statement and an audio prompt. The user's voice characteristics are encoded in the `speakerProfileId` property that's used for text to speech.

Once you have a personal voice, you can [use it](#) to synthesize speech in any of the 91 languages supported across 100+ locales. A locale tag isn't required. Personal voice uses automatic language detection at the sentence level. For more information, see [use personal voice in your application](#).

Tip

Check out the code samples in the [Speech SDK repository on GitHub](#)  to see how to use personal voice in your application.

Reference documentation

[Custom voice REST API reference documentation](#)

Responsible AI

We care about the people who use AI and the people who will be affected by it as much as we care about technology. For more information, see the Responsible AI [transparency notes](#).

Next steps

- [Create a project](#).
- Learn more about custom voice in the [overview](#).
- Learn more about Speech Studio in the [overview](#).

Create a project for personal voice

08/07/2025

Personal voice projects contain the user consent statement and the personal voice ID. You can only create a personal voice project using the custom voice API. You can't create a personal voice project in the Speech Studio.

Create a project

To create a personal voice project, use the [Projects_Create](#) operation of the custom voice API. Construct the request body according to the following instructions:

- Set the required `kind` property to `PersonalVoice`. The kind can't be changed later.
- Optionally, set the `description` property for the project description. The project description can be changed later.

Make an HTTP PUT request using the URI as shown in the following [Projects_Create](#) example.

- Replace `YourResourceKey` with your Speech resource key.
- Replace `YourResourceRegion` with your Speech resource region.
- Replace `ProjectId` with a project ID of your choice. The case sensitive ID must be unique within your Speech resource. The ID will be used in the project's URI and can't be changed later.

Azure CLI

```
curl -v -X PUT -H "Ocp-Apim-Subscription-Key: YourResourceKey" -H "Content-Type: application/json" -d '{
  "description": "Project description",
  "kind": "PersonalVoice"
}' "https://YourResourceRegion.api.cognitive.microsoft.com/customvoice/projects/ProjectId?api-version=2024-02-01-preview"
```

You should receive a response body in the following format:

JSON

```
{
  "id": "ProjectId",
  "description": "Project description",
  "kind": "PersonalVoice",
  "createdDateTime": "2024-09-01T05:30:00.000Z"
}
```

You use the project `id` in subsequent API requests to [add user consent](#) and [get a speaker profile ID](#).

Next steps

Add user consent to the personal voice project.

Add user consent to the personal voice project

08/07/2025

With the personal voice feature, it's required that every voice be created with explicit consent from the user. A recorded statement from the user is required acknowledging that the customer (Azure AI Speech resource owner) will create and use their voice.

To add user consent to the personal voice project, you provide the prerecorded consent audio file [from a publicly accessible URL \(Consents_Create\)](#) or [upload the audio file \(Consents_Post\)](#).

Consent statement

You need an audio recording of the user speaking the consent statement.

You can get the consent statement text for each locale from the text to speech GitHub repository. See [verbal-statement-all-locales.txt](#) for the consent statement. Below is a sample for the `en-US` locale:

```
"I [state your first and last name] am aware that recordings of my voice will be used by [state the name of the company] to create and use a synthetic version of my voice."
```

Supported audio formats for consent audio

See the table below for the supported formats for consent audio files:

 Expand table

Format	Sample rate	Bit rate	Bit depth
mp3	16 kHz, 24 kHz, 44.1 kHz, 48 kHz	128 kbps, 192 kbps, 256 kbps, 320 kbps	/
wav	16 kHz, 24 kHz, 44.1 kHz, 48 kHz	/	16-bit, 24-bit, 32-bit

Add consent from a file

In this scenario, the audio files must be available locally.

To add consent to a personal voice project from a local audio file, use the `Consents_Post` operation of the custom voice API. Construct the request body according to the following instructions:

- Set the required `projectId` property. See [create a project](#).
- Set the required `voiceTalentName` property. The voice talent name can't be changed later.
- Set the required `companyName` property. The company name can't be changed later.
- Set the required `audiodata` property with the consent audio file.
- Set the required `locale` property. This should be the locale of the consent. The locale can't be changed later. You can find the text to speech locale list [here](#).

Make an HTTP POST request using the URI as shown in the following `Consents_Post` example.

- Replace `YourResourceKey` with your Speech resource key.
- Replace `YourResourceRegion` with your Speech resource region.
- Replace `JessicaConsentId` with a consent ID of your choice. The case sensitive ID will be used in the consent's URI and can't be changed later.

Azure CLI

```
curl -v -X POST -H "Ocp-Apim-Subscription-Key: YourResourceKey" -F
'description="Consent for Jessica voice"' -F 'projectId="ProjectId"' -F
'voiceTalentName="Jessica Smith"' -F 'companyName="Contoso"' -F
'audiodata=@"D:\PersonalVoiceTest\jessica-consent.wav"' -F 'locale="en-US"'
"https://YourResourceRegion.api.cognitive.microsoft.com/customvoice/consents/Jessi
caConsentId?api-version=2024-02-01-preview"
```

You should receive a response body in the following format:

JSON

```
{
  "id": "JessicaConsentId",
  "description": "Consent for Jessica voice",
  "projectId": "ProjectId",
  "voiceTalentName": "Jessica Smith",
  "companyName": "Contoso",
  "locale": "en-US",
  "status": "NotStarted",
  "createdDateTime": "2024-09-01T05:30:00.000Z",
  "lastActionDateTime": "2024-09-02T10:15:30.000Z"
}
```

The response header contains the `Operation-Location` property. Use this URI to get details about the `Consents_Post` operation. Here's an example of the response header:

HTTP

```
Operation-Location:
https://eastus.api.cognitive.microsoft.com/customvoice/operations/070f7986-ef17-41d0-ba2b-907f0f28e314?api-version=2024-02-01-preview
Operation-Id: 070f7986-ef17-41d0-ba2b-907f0f28e314
```

Add consent from a URL

In this scenario, the audio files must already be stored in an Azure Blob Storage container.

To add consent to a personal voice project from the URL of an audio file, use the `Consents_Create` operation of the custom voice API. Construct the request body according to the following instructions:

- Set the required `projectId` property. See [create a project](#).
- Set the required `voiceTalentName` property. The voice talent name can't be changed later.
- Set the required `companyName` property. The company name can't be changed later.
- Set the required `audioUrl` property. The URL of the voice talent consent audio file. Use a URI with the [shared access signatures \(SAS\)](#) token.
- Set the required `locale` property. This should be the locale of the consent. The locale can't be changed later. You can find the text to speech locale list [here](#).

Make an HTTP PUT request using the URI as shown in the following `Consents_Create` example.

- Replace `YourResourceKey` with your Speech resource key.
- Replace `YourResourceRegion` with your Speech resource region.
- Replace `JessicaConsentId` with a consent ID of your choice. The case sensitive ID will be used in the consent's URI and can't be changed later.

Azure CLI

```
curl -v -X PUT -H "Ocp-Apim-Subscription-Key: YourResourceKey" -H "Content-Type: application/json" -d '{
  "description": "Consent for Jessica voice",
  "projectId": "ProjectId",
  "voiceTalentName": "Jessica Smith",
  "companyName": "Contoso",
  "audioUrl": "https://contoso.blob.core.windows.net/public/jessica-consent.wav?mySasToken",
  "locale": "en-US"
} '
```

```
"https://YourResourceRegion.api.cognitive.microsoft.com/customvoice/consents/JessicaConsentId?api-version=2024-02-01-preview"
```

You should receive a response body in the following format:

JSON

```
{
  "id": "JessicaConsentId",
  "description": "Consent for Jessica voice",
  "projectId": "ProjectId",
  "voiceTalentName": "Jessica Smith",
  "companyName": "Contoso",
  "locale": "en-US",
  "status": "NotStarted",
  "createdDateTime": "2024-09-01T05:30:00.000Z",
  "lastActionDateTime": "2024-09-02T10:15:30.000Z"
}
```

The response header contains the `Operation-Location` property. Use this URI to get details about the `Consents_Create` operation. Here's an example of the response header:

HTTP

```
Operation-Location:
https://eastus.api.cognitive.microsoft.com/customvoice/operations/070f7986-ef17-41d0-ba2b-907f0f28e314?api-version=2024-02-01-preview
Operation-Id: 070f7986-ef17-41d0-ba2b-907f0f28e314
```

Next steps

Create a personal voice.

Get a speaker profile ID for the personal voice

08/07/2025

To use personal voice in your application, you need to get a speaker profile ID. The speaker profile ID is used to generate synthesized audio with the text input provided.

You create a speaker profile ID based on the speaker's verbal consent statement and an audio prompt (a clean human voice sample between 5 - 90 seconds). The user's voice characteristics are encoded in the `speakerProfileId` property that's used for text to speech. For more information, see [use personal voice in your application](#).

ⓘ **Note**

The personal voice ID and speaker profile ID aren't same. You can choose the personal voice ID, but the speaker profile ID is generated by the service. The personal voice ID is used to manage the personal voice. The speaker profile ID is used for text to speech.

You provide the audio files [from a publicly accessible URL](#) (`PersonalVoices_Create`) or [upload the audio files](#) (`PersonalVoices_Post`).

Prompt audio format

The supported formats for prompt audio files are:

[Expand table](#)

Format	Sample rate	Bit rate	Bit depth
mp3	16 kHz, 24 kHz, 44.1 kHz, 48 kHz	128 kbps, 192 kbps, 256 kbps, 320 kbps	/
wav	16 kHz, 24 kHz, 44.1 kHz, 48 kHz	/	16-bit, 24-bit, 32-bit

Create personal voice from a file

In this scenario, the audio files must be available locally.

To create a personal voice and get the speaker profile ID, use the [PersonalVoices_Post](#) operation of the custom voice API. Construct the request body according to the following instructions:

- Set the required `projectId` property. See [create a project](#).
- Set the required `consentId` property. See [add user consent](#).
- Set the required `audiodata` property. You can specify one or more audio files in the same request.

Make an HTTP POST request using the URI as shown in the following [PersonalVoices_Post](#) example.

- Replace `YourResourceKey` with your Speech resource key.
- Replace `YourResourceRegion` with your Speech resource region.
- Replace `JessicaPersonalVoiceId` with a personal voice ID of your choice. The case sensitive ID will be used in the personal voice's URI and can't be changed later.

Azure CLI

```
curl -v -X POST -H "Ocp-Apim-Subscription-Key: YourResourceKey" -F
'projectId="ProjectId"' -F 'consentId="JessicaConsentId"' -F
'audiodata=@"D:\PersonalVoiceTest\CNVSample001.wav"' -F
'audiodata=@"D:\PersonalVoiceTest\CNVSample002.wav"' "
https://YourResourceRegion.api.cognitive.microsoft.com/customvoice/personalvoices/
JessicaPersonalVoiceId?api-version=2024-02-01-preview"
```

You should receive a response body in the following format:

JSON

```
{
  "id": "JessicaPersonalVoiceId",
  "speakerProfileId": "3059912f-a3dc-49e3-bdd0-02e449df1fe3",
  "projectId": "ProjectId",
  "consentId": "JessicaConsentId",
  "status": "NotStarted",
  "createdDateTime": "2024-09-01T05:30:00.000Z",
  "lastActionDateTime": "2024-09-02T10:15:30.000Z"
}
```

Use the `speakerProfileId` property to integrate personal voice in your text to speech application. For more information, see [use personal voice in your application](#).

The response header contains the `Operation-Location` property. Use this URI to get details about the [PersonalVoices_Post](#) operation. Here's an example of the response header:

Operation-Location:

`https://eastus.api.cognitive.microsoft.com/customvoice/operations/1321a2c0-9be4-471d-83bb-bc3be4f96a6f?api-version=2024-02-01-preview`

Operation-Id: 1321a2c0-9be4-471d-83bb-bc3be4f96a6f

Create personal voice from a URL

In this scenario, the audio files must already be stored in an Azure Blob Storage container.

To create a personal voice and get the speaker profile ID, use the [PersonalVoices_Create](#) operation of the custom voice API. Construct the request body according to the following instructions:

- Set the required `projectId` property. See [create a project](#).
- Set the required `consentId` property. See [add user consent](#).
- Set the required `audios` property. Within the `audios` property, set the following properties:
 - Set the required `containerUrl` property to the URL of the Azure Blob Storage container that contains the audio files. Use [shared access signatures \(SAS\) SAS for a container](#) with both read and list permissions.
 - Set the required `extensions` property to the extensions of the audio files.
 - Optionally, set the `prefix` property to set a prefix for the blob name.

Make an HTTP PUT request using the URI as shown in the following [PersonalVoices_Create](#) example.

- Replace `YourResourceKey` with your Speech resource key.
- Replace `YourResourceRegion` with your Speech resource region.
- Replace `JessicaPersonalVoiceId` with a personal voice ID of your choice. The case sensitive ID will be used in the personal voice's URI and can't be changed later.

Azure CLI

```
curl -v -X PUT -H "Ocp-Apim-Subscription-Key: YourResourceKey" -H "Content-Type: application/json" -d '{
  "projectId": "ProjectId",
  "consentId": "JessicaConsentId",
  "audios": {
    "containerUrl": "https://contoso.blob.core.windows.net/voicecontainer?mySasToken",
    "prefix": "jessica/",
    "extensions": [
      ".wav"
    ]
  }
}
```

```

    ]
  }
} '
"https://YourResourceRegion.api.cognitive.microsoft.com/customvoice/personalvoices
/JessicaPersonalVoiceId?api-version=2024-02-01-preview"

# Ensure the `containerUrl` has both read and list permissions.
# Ensure the `.wav` files are located in the "jessica" folder within the
container. The `prefix` matches all `.wav` files in the "jessica" folder. If there
is no such folder, the prefix will match `.wav` files with names starting with
"jessica".

```

You should receive a response body in the following format:

JSON

```

{
  "id": "JessicaPersonalVoiceId",
  "speakerProfileId": "3059912f-a3dc-49e3-bdd0-02e449df1fe3",
  "projectId": "ProjectId",
  "consentId": "JessicaConsentId",
  "status": "NotStarted",
  "createdDateTime": "2024-09-01T05:30:00.000Z",
  "lastActionDateTime": "2024-09-02T10:15:30.000Z"
}

```

Use the `speakerProfileId` property to integrate personal voice in your text to speech application. For more information, see [use personal voice in your application](#).

The response header contains the `Operation-Location` property. Use this URI to get details about the [PersonalVoices_Create](#) operation. Here's an example of the response header:

HTTP

```

Operation-Location:
https://eastus.api.cognitive.microsoft.com/customvoice/operations/1321a2c0-9be4-
471d-83bb-bc3be4f96a6f?api-version=2024-02-01-preview
Operation-Id: 1321a2c0-9be4-471d-83bb-bc3be4f96a6f

```

Next steps

Use personal voice in your application.

Use personal voice in your application

08/07/2025

You can use the [speaker profile ID](#) for your personal voice to synthesize speech in any of the 91 languages supported across 100+ locales. A locale tag isn't required. Personal voice uses automatic language detection at the sentence level.

Integrate personal voice in your application

You need to use [speech synthesis markup language \(SSML\)](#) to use personal voice in your application. SSML is an XML-based markup language that provides a standard way to mark up text for the generation of synthetic speech. SSML tags are used to control the pronunciation, volume, pitch, rate, and other attributes of the speech synthesis output.

- The `speakerProfileId` property in SSML is used to specify the [speaker profile ID](#) for the personal voice.
- The voice name is specified in the `name` property in SSML. For personal voice, the voice name must be one of the supported base model voice names. To get a list of supported base model voice names, use the [BaseModels_List](#) operation of the custom voice API.

! Note

The voice names labeled with the `Latest`, such as `DragonLatestNeural` or `PhoenixLatestNeural`, will be updated from time to time; its performance may vary with updates for ongoing improvements. If you would like to use a fixed version, select one labeled with a version number, such as `PhoenixV2Neural`.

- `DragonLatestNeural` is a base model with superior voice cloning similarity compared to `PhoenixLatestNeural`. `PhoenixLatestNeural` is a base model with more accurate pronunciation and lower latency than `DragonLatestNeural`.
- For personal voice, you can use the `<lang xml:lang>` element to adjust the speaking language. It's the same as with multilingual voices. See [how to use the lang element to speak different languages](#).

Here's example SSML in a request for text to speech with the voice name and the speaker profile ID. The sample also demonstrates how to switch languages from `en-US` to `zh-HK` using the `<lang xml:lang>` element.

```
<speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis'
xmlns:mstts='http://www.w3.org/2001/mstts' xml:lang='en-US'>
  <voice name='DragonLatestNeural'>
    <mstts:ttseembedding speakerProfileId='your speaker profile ID here'>
      I'm happy to hear that you find me amazing and that I have made your
      trip planning easier and more fun.
      <lang xml:lang='zh-HK'>我很高興聽到你覺得我很了不起，我讓你的旅行計劃更輕
      鬆、更有趣。</lang>
    </mstts:ttseembedding>
  </voice>
</speak>
```

You can use the SSML via the [Speech SDK](#) or [REST API](#).

- **Real-time speech synthesis:** Use the [Speech SDK](#) or [REST API](#) to convert text to speech.
 - When you use Speech SDK, don't set Endpoint ID, just like prebuild voice.
 - When you use REST API, please use standard voices endpoint.

Supported and unsupported SSML elements for personal voice

For detailed information on the supported and unsupported SSML elements for Phoenix and Dragon models, refer to the following table. For instructions on how to use SSML elements, refer to the [SSML document structure and events](#).

 Expand table

Element	Description	Supported in Phoenix	Supported in Dragon
<voice>	Specifies the voice and optional effects (eq_car and eq_telecomhp8k).	Yes	Yes
<mstts:express-as>	Specifies speaking styles and roles.	No	No
<mstts:ttseembedding>	Specifies the speakerProfileId property for a personal voice.	Yes	Yes
<lang xml:lang>	Specifies the speaking language.	Yes	Yes
<prosody>	Adjusts pitch, contour, range, rate, and volume.		
pitch	Indicates the baseline pitch for the text.	No	No

Element	Description	Supported in Phoenix	Supported in Dragon
<code>contour</code>	Represents changes in pitch.	No	No
<code>range</code>	Represents the range of pitch for the text.	No	No
<code>rate</code>	Indicates the speaking rate of the text.	Yes	Yes
<code>volume</code>	Indicates the volume level of the speaking voice.	No	No
<code><emphasis></code>	Adds or removes word-level stress for the text.	No	No
<code><audio></code>	Embeds prerecorded audio into an SSML document.	Yes	No
<code><mstts:audioduration></code>	Specifies the duration of the output audio.	No	No
<code><mstts:backgroundaudio></code>	Adds background audio to your SSML documents or mixes an audio file with text to speech.	Yes	No
<code><phoneme></code>	Specifies phonetic pronunciation in SSML documents.		
<code>ipa</code>	One of the phonetic alphabets.	Yes	No
<code>sapi</code>	One of the phonetic alphabets.	No	No
<code>ups</code>	One of the phonetic alphabets.	Yes	No
<code>x-sampa</code>	One of the phonetic alphabets.	Yes	No
<code><lexicon></code>	Defines how multiple entities are read in SSML.	Yes	Yes (only support alias)
<code><say-as></code>	Indicates the content type, such as number or date, of the element's text.	Yes	Yes
<code><sub></code>	Indicates that the alias attribute's text value should be pronounced instead of the element's enclosed text.	Yes	Yes
<code><math></code>	Uses the MathML as input text to properly pronounce mathematical notations in the output audio.	Yes	No
<code><bookmark></code>	Gets the offset of each marker in the audio stream.	Yes	No

Element	Description	Supported in Phoenix	Supported in Dragon
<code><break></code>	Overrides the default behavior of breaks or pauses between words.	Yes	Yes
<code><mstts:silence></code>	Inserts pauses before or after text, or between two adjacent sentences.	Yes	No
<code><mstts:viseme></code>	Defines the position of the face and mouth while a person is speaking.	Yes	No
<code><p></code>	Denotes paragraphs in SSML documents.	Yes	Yes
<code><s></code>	Denotes sentences in SSML documents.	Yes	Yes

Supported and unsupported SDK features for personal voice

The following table outlines which SDK features are supported for Phoenix and Dragon models. For details on how to utilize these SDK features in your applications, refer to [Subscribe to synthesizer events](#).

 Expand table

SDK features	Description	Supported in Phoenix	Supported in Dragon
Word boundary	Signals that a word boundary was received during synthesis, providing precise word timing during the speech synthesis process.	Yes	No
Viseme events	Provides viseme (lips, jaw, and tongue movement) information during synthesis, allowing visual synchronization.	Yes	No

Reference documentation

[Custom voice REST API reference documentation](#)

Next steps

- Learn more about custom voice in the [overview](#).

- Learn more about Speech Studio in the [overview](#).

What is Text to speech avatar?

Text to speech avatar converts text into a digital video of a photorealistic human (either a standard avatar or a [custom text to speech avatar](#)) speaking with a natural-sounding voice. The text to speech avatar video can be synthesized asynchronously or in real time. Developers can build applications integrated with text to speech avatar through an API, or use a content creation tool on Speech Studio to create video content without coding.

With text to speech avatar's advanced neural network models, the feature empowers users to deliver life-like and high-quality synthetic talking avatar videos for various applications while adhering to [responsible AI practices](#).

Tip

To convert text to speech with a no-code approach, try the [Text to speech avatar tool in Speech Studio](#) .

Avatar capabilities

Text to speech avatar capabilities include:

- Converts text into a digital video of a photorealistic human speaking with natural-sounding voices powered by Azure AI text to speech.
- Provides a collection of standard avatars.
- Azure AI text to speech generates the voice of the avatar. For more information, see [Avatar voice and language](#).
- Synthesizes text to speech avatar video asynchronously with the [batch synthesis API](#) or in [real-time](#).
- Provides a [content creation tool](#)  in Speech Studio for creating video content without coding.
- Enables real-time avatar conversations through the [live chat avatar tool](#)  in Speech Studio.

With text to speech avatar's advanced neural network models, the feature empowers you to deliver lifelike and high-quality synthetic talking avatar videos for various applications while adhering to responsible AI practices.

Avatar voice and language

You can choose from a range of standard voices for the avatar. The language support for text to speech avatar is the same as the language support for text to speech. For details, see [Language and voice support for the Speech service](#). Standard text to speech avatars can be accessed through the [Speech Studio portal](#) or via API.

The voice in the synthetic video could be an Azure AI Speech standard voice or the [custom voice](#) of voice talent selected by you.

Avatar video output

Both batch synthesis and real-time synthesis resolution are 1920 x 1080, and the frames per second (FPS) are 25. Batch synthesis codec can be h264, hevc, or av1 if the format is `mp4` and can set codec as vp9 or av1 if the format is `webm`; only `vp9` can contain an alpha channel. Real-time synthesis codec is h264. Video bitrate can be configured for both batch synthesis and real-time synthesis in the request; the default value is 2000000; more detailed configurations can be found in the sample code.

[Expand table](#)

	Batch synthesis	Real-time synthesis
Resolution	1920 x 1080	1920 x 1080
FPS	25	25
Codec	h264/hevc/vp9/av1	h264

Custom text to speech avatar

You can create custom text to speech avatars that are unique to your product or brand. All it takes to get started is taking 10 minutes of video recordings. If you're also fine-tuning a professional voice for the actor, the avatar can be highly realistic.

Voice sync for avatar is trained alongside the custom avatar utilizing audio from the training video. The voice is exclusively associated with the custom avatar and can't be independently used.

[Professional voice fine-tuning](#) and [custom text to speech avatar](#) are separate features. You can use them independently or together. If you plan to also use [professional voice fine-tuning](#) with a text to speech avatar, you need to deploy or [copy](#) your fine-tuned professional voice model to one of the [avatar supported regions](#).

For more information, see [What is custom text to speech avatar](#).

Sample code

Sample code for text to speech avatar is available on [GitHub](#). These samples cover the most popular scenarios:

- [Batch synthesis \(REST\)](#)
- [Real-time synthesis \(SDK\)](#)
- [Live chat with Azure OpenAI in behind \(SDK\)](#)
- To create a live chat APP with Azure OpenAI [On Your Data](#), you can refer to [this sample code](#) (search "On Your Data")

Pricing

- Throughout an avatar real-time session or batch content creation, the text to speech, speech to text, Azure OpenAI, or other Azure services are charged separately.
- Voice sync for avatar (via custom avatar training) is charged the same as a personal voice in terms of voice creation and synthesis. The storage of the voice is free.
- Refer to [text to speech avatar pricing note](#) to learn how billing works for the text-to-speech avatar feature.
- For the detailed pricing, see [Speech service pricing](#). Avatar pricing is only visible for service regions where the feature is available. For the current list of supported regions, see the [Speech service regions table](#).

Available locations

For the current list of regions that support text to speech avatar, see the [Speech service regions table](#).

Responsible AI

We care about the people who use AI and the people who will be affected by it as much as we care about technology. For more information, see the Responsible AI [transparency notes](#) and [disclosure for voice and avatar talent](#).

Next steps

- [Use batch synthesis for text to speech avatar](#)
 - [What is custom text to speech avatar](#)
-

Content credentials

08/07/2025

Content credentials help you verify that Azure text to speech avatar generated your video content. This transparency feature automatically adds tamper-evident information about your video's origin and creation history, following industry standards from the [Coalition for Content Provenance and Authenticity \(C2PA\)](#).

Use content credentials to:

- Confirm video authenticity for your audiences
- Meet transparency requirements for AI-generated content
- Build trust in your avatar-powered solutions

Content credentials are automatically applied to all supported video formats—no additional setup required.

What are content credentials?

Azure text to speech avatar adds a content credentials manifest to each avatar video, giving you information about the video's origin. A cryptographic signature secures the manifest, linking it back to Azure text to speech avatar.

The manifest has several key pieces of information:

 Expand table

Field name	Field content
"generator"	This field has a value of "Microsoft Azure Txt to Speech Avatar Service" for all applicable videos, confirming the AI-generated nature of the video.
"when"	The timestamp of when the content credentials were created.

Content credentials in Azure text to speech avatar can help people understand when video content is generated by the Azure text to speech avatar system. For more information on how to responsibly build solutions with text to speech avatar models, see the [Text to speech transparency note](#).

Limitations